Design Project 1 – 1P13 Robotic End Effector System

ENGINEER 1P13 – Integrated Cornerstone Design Projects in Engineering

Tutorial T07

Team Thur-04

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Submitted: December 3, 2025

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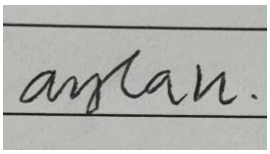
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Academic Integrity Statement

The student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University.

Ayla Nasir 400639418



_____
(Student Signature)

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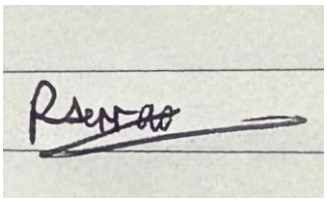
Wenyue Jiang 400626520



_____
(Student Signature)

The student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University.

Raphael Serrao 400626455



_____
(Student Signature)

The student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University.

Sherry Zhao 400633555



(Student Signature)

The student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University.

Shivank Nambiar 400628172



(Student Signature)

Executive Summary

Design Strategy and Concept Selection

Our project centered on creating a robotic end-effector and a matching software system that could work together to pick up and package objects of different shapes in a warehouse environment. From the beginning, we understood that our design needed to balance simplicity with precision. The Q-Arm had fixed limitations, especially its built-in safety feature that prevents a full closure if it senses resistance. Because of this, we had to design a mechanism that could still grip objects securely without relying on strong force. This guided almost every choice we made. We aimed for a design that was reliable, lightweight, and adaptable, while also keeping the software clear, organized, and easy to connect to the physical system.

When choosing our mechanism concept, we focused on what we could control: the material, the shape of the claws, the weight distribution, and how the parts interacted. We avoided overly complex ideas because adding gears and multiple moving parts would create more room for error. Instead, we chose a simple claw mechanism that aligned with our project goals [1]. A straightforward design meant fewer tolerances to worry about and a higher chance of consistent performance once the Q-Arm started moving. Our strategy for the software followed a similar idea—each function needed to be clean, modular, and easy to integrate. By starting with flowcharts, we made sure that the individual pieces would fit together just like the project required: authentication, scanning, product lookup, packing, and generating a final receipt.

Design Iteration and Refinement Process

Reaching our final design took many rounds of iteration. As a team, we began by sharing our sketches and CAD models and openly discussing the strengths and weaknesses of each. That conversation helped us merge our favorite features into one design. But once we began fabricating, new challenges appeared. Our gear-mounting hole did not fit onto the servo shaft because we underestimated how much 3D-printing tolerances matter. Fixing that meant adjusting our dimensions and reprinting the part. We also experimented with adding textured "jaws" to the claws, hoping the extra grip would help [2]. Instead, the objects slipped even more because both the claws and objects were printed in the same material. It wasn't until we spoke with our TA, who emphasized that friction mattered more than force, that we stopped trying to create complex gripping patterns and turned toward using rubber. Applying rubber bands onto the claws made a noticeable difference and became one of the most important refinements in our build [3].

Our software went through its own set of iterations. In our first draft, our flowcharts were missing barcode logic and didn't meet professional standards. After feedback, we refined them, so they clearly showed how information should move through each function. The coding stages brought new problems: reading from CSV files, formatting invoices so the numbers aligned, and handling unexpected input. The hardest part, however, was determining the Q-Arm coordinates. Even a small mistake caused the arm to bump into nearby objects or miss the intended pickup point completely. It took a lot of trial and error, but eventually we were able to create motions that were reliable enough for testing. When all the functions came together, we still had to fix mismatched return types and formatting issues, but by the end, the full workflow ran smoothly.

Testing Results and System Performance

When we tested the fully integrated system, the software performed consistently well. Users could sign up and authenticate, products scanned correctly, and the program generated clean, readable invoices with the correct discount and tax calculations. The customer summary also showed accurate totals. The mechanical performance, however, was more unpredictable. While the rubber covered claws picked up several objects successfully, the Q-Arm's safety feature limited how tightly the mechanism could close. This means that smaller or smoother objects sometimes slip. There were also occasional collisions when the arm tried to grab objects placed close together. On top of that, the Q-Arm did not always repeat the same motion perfectly, even

with identical code. Still, the system managed to complete the required workflow in several trials, showing partial but meaningful success.

Future improvements and Next Steps

If we were to continue this project, several refinements would help improve performance. Mechanically, we would redesign the claw tips to better match the shapes of the objects, which would allow the arm to establish a more secure grip even without full closure. We would also explore more textured or higher-friction materials—such as silicone—to replace the rubber [4]. Adjusting tolerances around the servo shaft would make motion smoother and reduce unwanted wobbling. On the software side, adding automated calibration, improving error handling, and increasing buffer distances during movement would help prevent collisions and make the system more consistent.

Reflection on Project Timeline and Collaboration

Reflecting on our initial Gantt chart, our actual progress differed from what we planned. Issues like reprinting parts, adjusting tolerances, and discovering the Q-Arm's safety limitation all happened later than expected. This pushed back our testing schedule, and by the time we tried to book extra Q-Arm sessions, most slots were already full, even though this was still over a week before the deadline. Scheduling challenges also played a role, since it wasn't always possible for all of us to meet outside of the design studio. As a result, some problems took longer to fix because we had to wait for the next studio session. From this experience, we learned how important it is to build extra buffer time into a schedule. For future projects, we would try to work at least one or two weeks ahead, start integration earlier, and secure testing time sooner.

Overall, the project taught us that even with careful planning, unexpected issues can appear at any stage. What matters most is how quickly a team can recognize the problem, adjust, and keep moving forward. Our final design reflects that process.

Reference List

[1] "Types of end-effectors: Essential tools for industrial automation - Standard Bots," *standardbots.com*, Sep. 24, 2025. https://standardbots.com/blog/types-of-end-effectors-essential-tools-for-industrial-automation

[2] E. Newton, "Everything You Need to Know About End-of-Arm Tooling," *Robot Shop*, Dec. 13, 2021. https://community.robotshop.com/blog/show/everything-you-need-to-know-about-end-of-arm-tooling-1

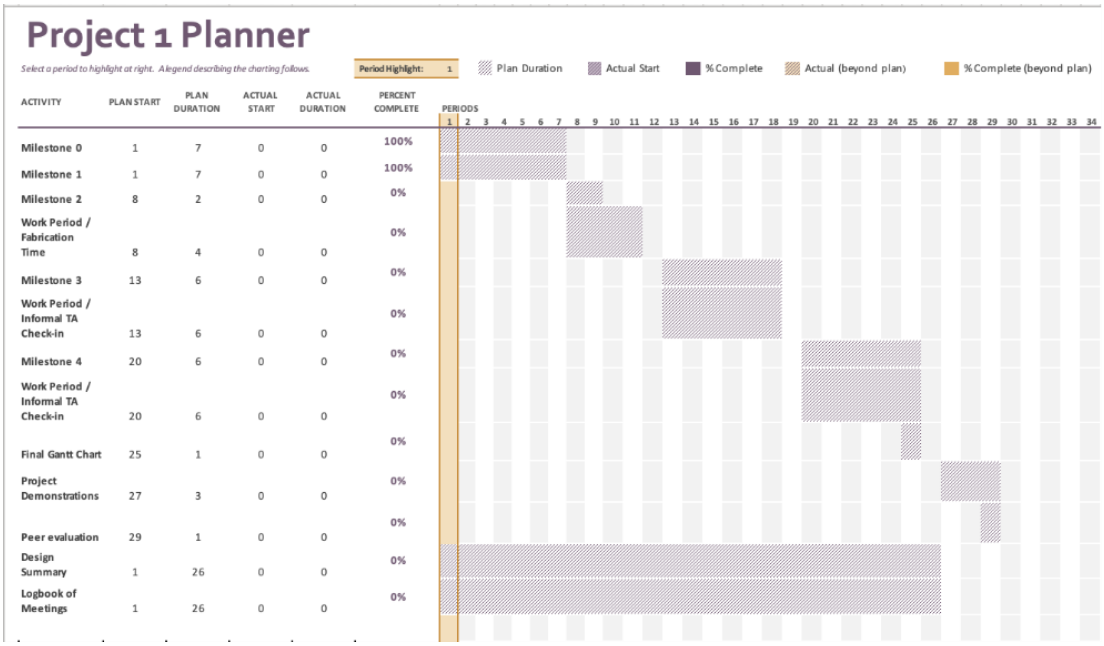
[3] rggroup, "End Effector Design Factors | Robotics," *RG Group*, Feb. 13, 2019. https://www.rg-group.com/end-effector-design-factors/

[4] I. PFA, "PFA EOAT | Robotic Grippers, Gripper Pads, RCC, OPD," *PFA, Inc.*, Sep. 10, 2022. https://www.pfa-inc.com/modular-end-of-arm-automation/gripper-pads/#gripper-pads-overview

Appendices

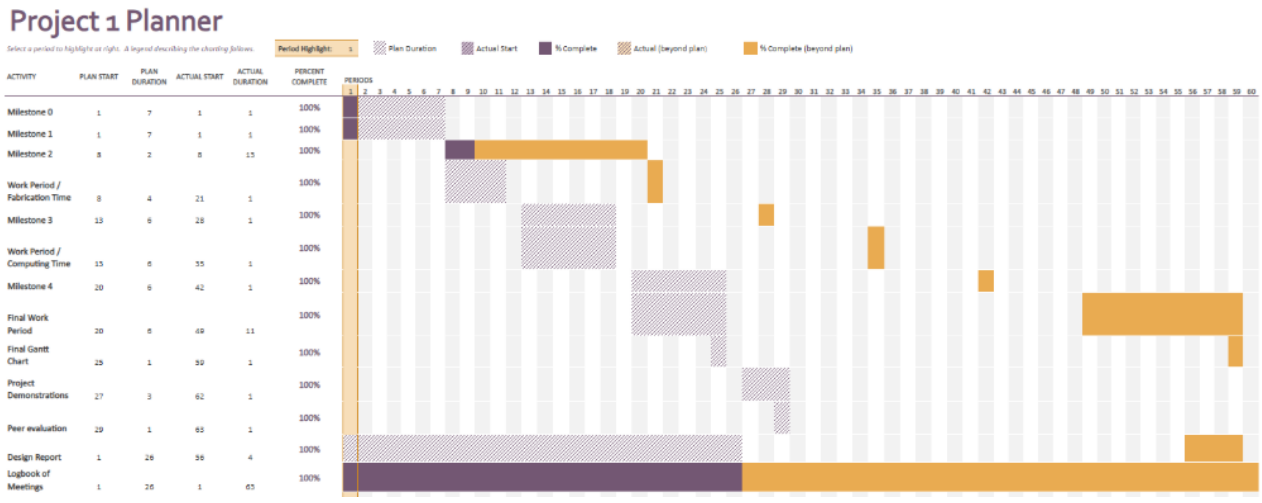
Appendix A: Project Schedule

Preliminary Gantt Chart



ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Milestone 0	1	7	0	0	100%
Milestone 1	1	7	0	0	100%
Milestone 2	8	2	0	0	0%
Work Period / Fabrication Time	8	4	0	0	0%
Milestone 3	13	6	0	0	0%
Work Period / Informal TA Check-in	13	6	0	0	0%
Milestone 4	20	6	0	0	0%
Work Period / Informal TA Check-in	20	6	0	0	0%
Final Gantt Chart	25	1	0	0	0%
Project Demonstrations	27	3	0	0	0%
Peer evaluation	29	1	0	0	0%
Design Summary	1	26	0	0	0%
Logbook of Meetings	1	26	0	0	0%

Final Gantt Chart



ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Milestone 0	1	7	1	1	100%
Milestone 1	1	7	1	1	100%
Milestone 2	8	2	8	15	100%
Work Period / Fabrication Time	8	4	21	1	100%
Milestone 3	13	6	28	1	100%
Work Period / Computing Time	13	6	35	1	100%
Milestone 4	20	6	42	1	100%
Final Work Period	20	6	49	11	100%
Final Gantt Chart	25	1	59	1	100%
Project Demonstrations	27	3	62	1	100%
Peer evaluation	29	1	65	1	100%
Design Report	1	26	36	4	100%
Logbook of Meetings	1	26	1	65	100%

Logbook of Additional Meetings and Discussions

Time	Content

Oct 21st 6:30-8:30pm	Finishing Milestone 2 (stage 2). Half of the group members worked on CAD prototype drawings; the rest worked on justifications, including justification of the reason we chose the prototype as well as further optimizations.
Nov 24th 3:30-5:30pm	Based on the performance of our end effector and the fact that it cannot pick up anything more than level 1, we concluded that our prototype of chipped claw does not add much of the friction to pick the designated objects up. To improve this, we decided to meet up, reprint the end effector, add rubber bands on the claws to increase the grid, as well as combine the codes and test runs.
Dec 1st 2:30-4:30pm	Finalize report and code. We solved some common errors with the code including the lack of exceptions, while loop errors as well as userid existent check. Finishing Engineering Drawing with proper layout, scale and dimensions.

Appendix B: Scheduled Weekly Meetings

ENGINEER 1P13

MEETING WITH TEAM 04-THURSDAY, OCT 2, OCT 9, OCT 23, OCT 30, NOV 6, NOV 13, 2025

Attendance

Role	Name	Mac ID	Attendance (Yes/No)
Manager	Ayla Nasir	nasira20	Yes
Administrator	Wenyue Jiang	jiangw74	Yes
Coordinator	Sherry Zhao	zhaot75	Yes
Subject Matter Expert	Shivank Sunil Nambiar	nambis2	Yes
Subject Matter Expert	Raphael Serrao	serrar2	Yes

Agenda Items

1. • Brainstorm functions, objectives, and constraints for Milestone 1
2. • Share early concept ideas (before doing individual sketches)
3. • Finish sketches + refine initial problem framing

Meeting Minutes

1. Week 5, Milestone 1 (Stage 2)

   Feedback:

   a. -Overall: should have more formal writing, no personal pronouns, use verbs instead
   b. -Function part: N/A
   c. -Constraint: Combine with safety purposes
   d. -Objective: Injury concerns more fits in constraint part
2. Week 6, Milestone 2 (Stage 3)

   Feedback:

   a. -The sketch was in the iam file, but it should be in the ipt file
   b. -Spur gear should be in ipt file
   c. -Enlarge the 2 parts in the gear feature because the 2 gears are not connecting and working with one another to move
   d. -Extend the arms so they can properly pick up items, right now the arms are too thin and small
   e. -Use calliper for accurate measurements to increase reliability of parts to be the proper size
3. Week 7, Work Period: Fabrication Time (STAGE 3)

Feedback:

a. -The sketch was in the iam file, but it should be in the ipt file
b. -Spur gear should be in ipt file
c. -Enlarge the 2 parts in the gear feature because the 2 gears are not connecting and working with one another to move
d. -Extend the arms so they can properly pick up items, right now the arms are too thin and small
e. -Use calliper for accurate measurements to increase reliability of parts to be the proper size

4. Week 8, Milestone 3 (Stage 3)

   Feedback:

   a. -The flowchart should include the standard flowchart symbols
   b. -In the future, a software could be used to make flowchart look more professional
   c. -New feedback (keep in mind as we are coding and moving forward):
   d. -We didn't mention the scanned barcode
   e. -Scan one --> check product, go through full loop, then check if the user wants to scan again

5. Week 9, Work Period: Code Your Function (Stage 1)

   Feedback:

   a. -Ensure the Q-Arm code for any movement is properly implemented into the pack_products(): function
   b. -Main(): function may have to be refined as we implement the rest of our individual code

6. Week 10, Milestone 4 (Stage 3)

   Feedback:

   a. -The positioning of the Q-Arm (where it moves) needs to be more accurate and efficient
   b. -The final code should follow a logical order of when the functions are called (eg. sign_up(): function should be around the top of the program code, whereas main(): would be near the end)
   c. -The scan_barcode() code provided needs to be implemented in the python shell, and should be able to retrieve the name of the products

Post-Meeting Action Items

1. Week 5, Milestone 1 (Stage 2)

   -Adjust overall vocabulary to fit professional environment

   -Keeping safety as a priority going forward while coming up with ideas

2. Week 6, Milestone 2 (Stage 3)

   -Measuring tools in inventor will be used to ensure parts work in harmony and meet their role in the mechanism

-The arms will be extended and thicken so they can properly pick up the items

-The shape of the claws will be changed from curve spurs to flatter spurs to be more reliable and versatile

-The area of the spurs around the claws will also be increased to pick up the level 4 design

3. Week 7, Work Period: Fabrication Time (STAGE 3)
   -Measuring tools in inventor will be used to ensure parts work in harmony and meet their role in the mechanism

   -The arms will be extended and thicken so they can properly pick up the items

   -The shape of the claws will be changed from curve spurs to flatter spurs to be more reliable and versatile

   -The area of the spurs around the claws will also be increased to pick up the level 4 design
4. Week 8, Milestone 3 (Stage 3)
   - In the future, a flowchart software will be used to ensure the flowchart has the correct symbols, and looks professional
5. Week 9, Work Period: Code Your Function (Stage 1)
   - Ensure the Q-Arm code for any movement is properly implemented into the pack_products(): function

   - Main(): function may have to be refined as we implement the rest of our individual code
6. Week 10, Milestone 4 (Stage 3)
   - The code for more precise and correct positioning of the Q-Arm to relocate each object will be properly implemented
   - When the individual codes and any other required code are placed together, it will follow a logical flow
   - The scan_barcode() function will be implemented and tested to ensure that it works correctly.

Appendix C: Comprehensive List of Sources

Source Materials Database

[1] A. Sharma, "Design Study of End Effectors," Feb. 2015. Available: https://www.ijeat.org/wp-content/uploads/papers/v4i3/C3789024315.pdf

[2] E. Newton, "Everything You Need to Know About End-of-Arm Tooling," Robot Shop, Dec. 13, 2021. https://community.robotshop.com/blog/show/everything-you-need-to-know-about-end-of-arm-tooling-1

[3] I. PFA, "PFA EOAT | Robotic Grippers, Gripper Pads, RCC, OPD," PFA, Inc., Sep. 10, 2022. https://www.pfa-inc.com/modular-end-of-arm-automation/gripper-pads/#gripper-pads-overview

[4] "10 Important Features of High Tack Application Tape - MPI Matco," MPI Matco, Sep. 03, 2018. Available: https://mpi-matco.com/manufacturing/10-important-features-high-tack-application-tape/

[5] rggroup, "End Effector Design Factors | Robotics," RG Group, Feb. 13, 2019. https://www.rg-group.com/end-effector-design-factors/

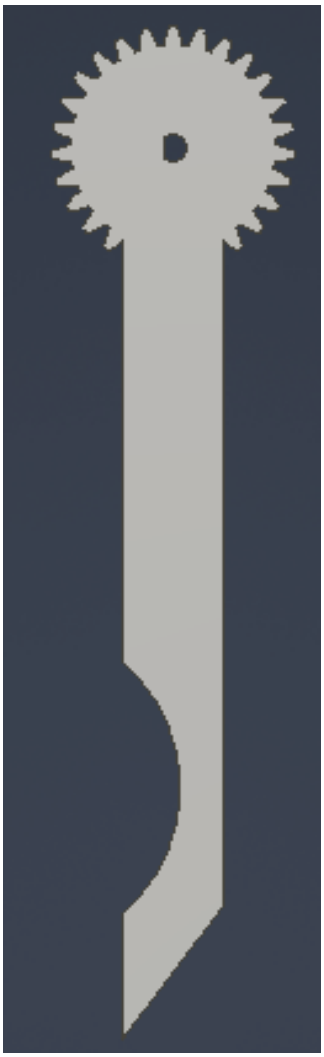
[6] Rubber bands. VEX Robotics. Accessed: Nov. 20, 2025. [Online]. Available: https://www.vexrobotics.com/rubberbands.html

[7] "Types of end-effectors: Essential tools for industrial automation - Standard Bots," standardbots.com, Sep. 24, 2025. https://standardbots.com/blog/types-of-end-effectors-essential-tools-for-industrial-automation

Appendix D: Additional Documentation

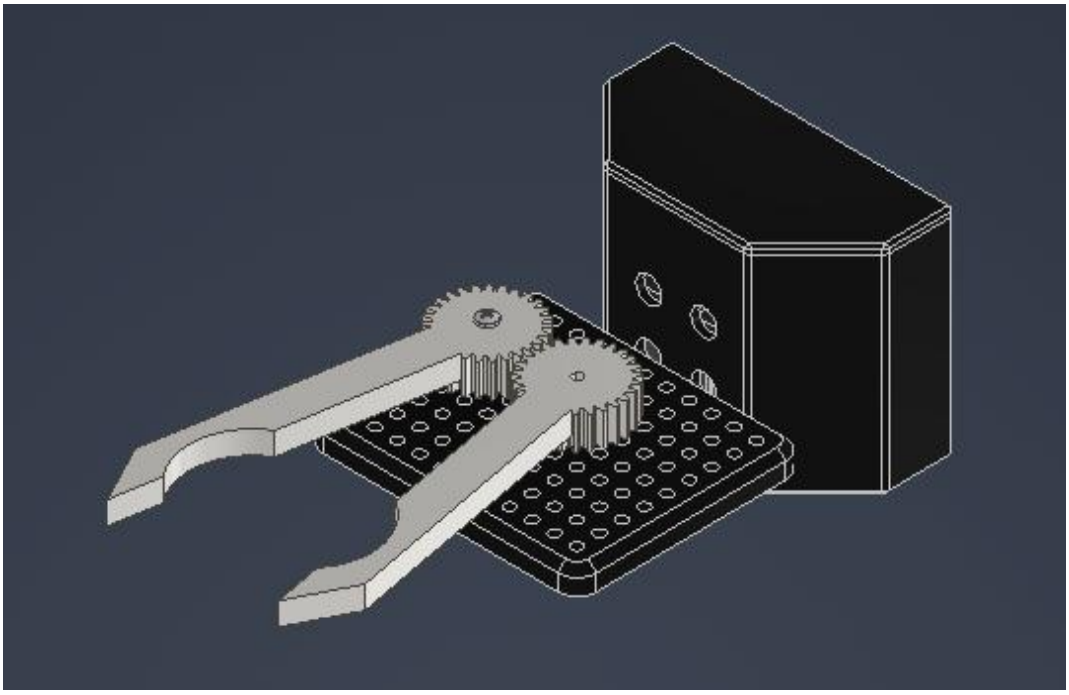
Screenshots of each solid model part:

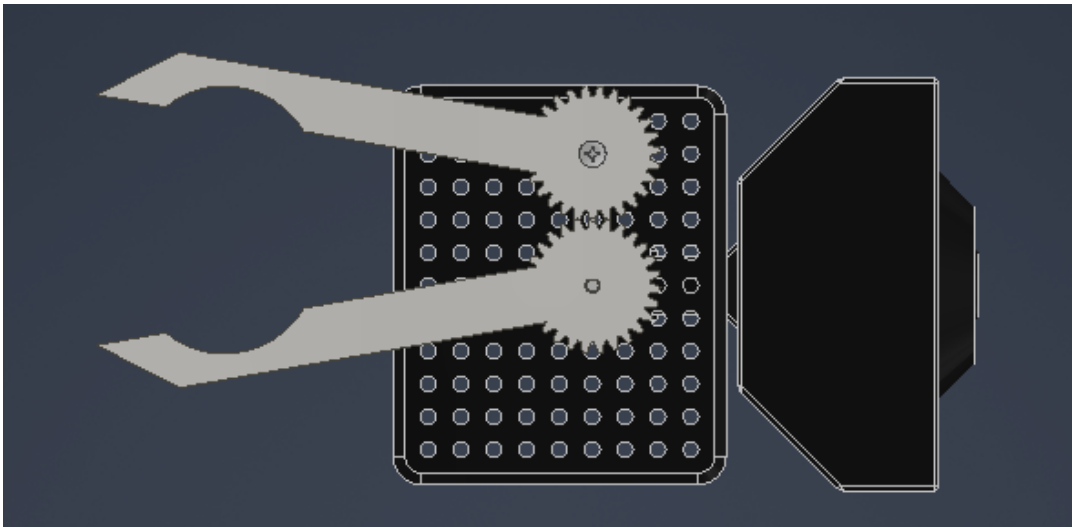


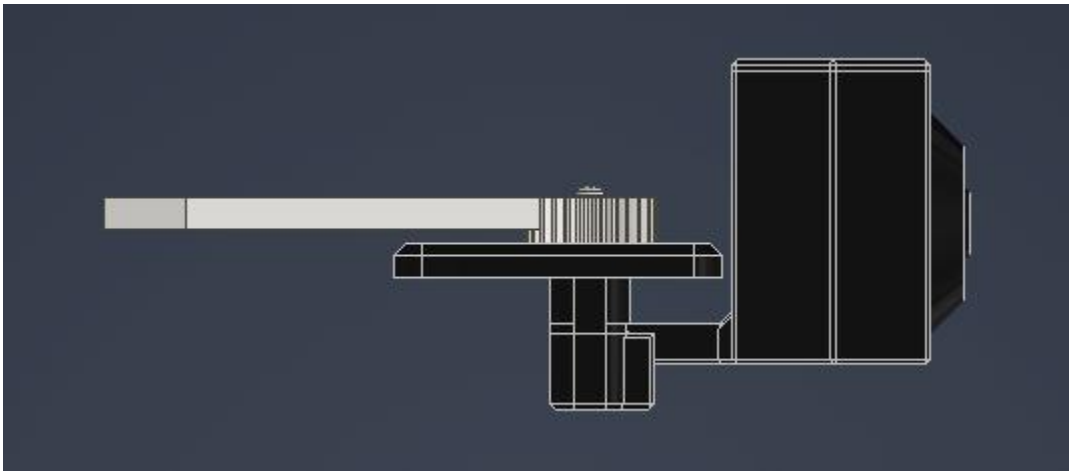


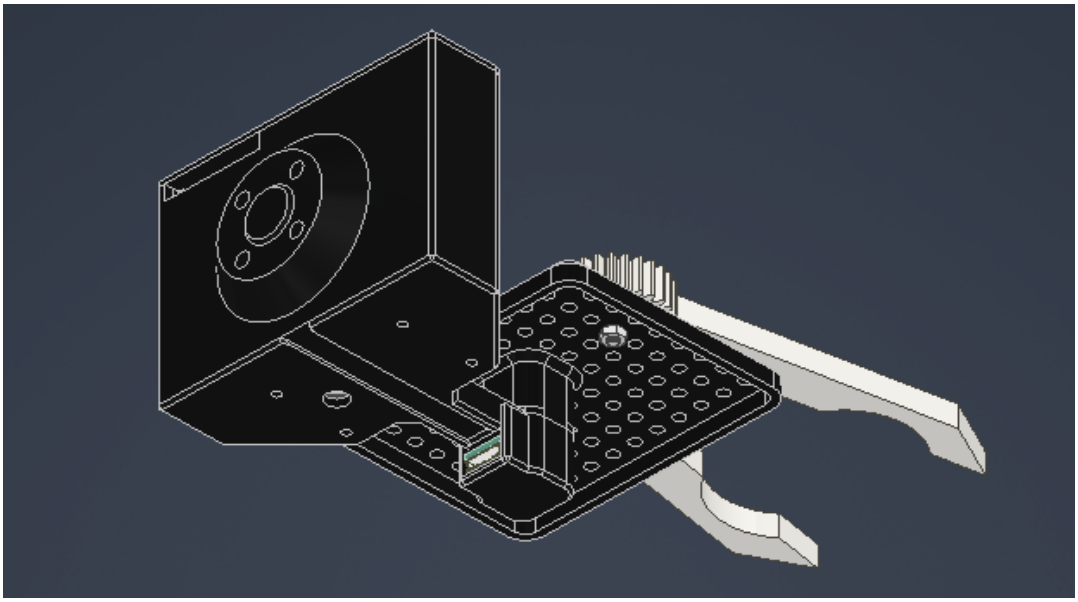


Screenshots of overall mechanism (including the platform and any fasteners used), and computer program:







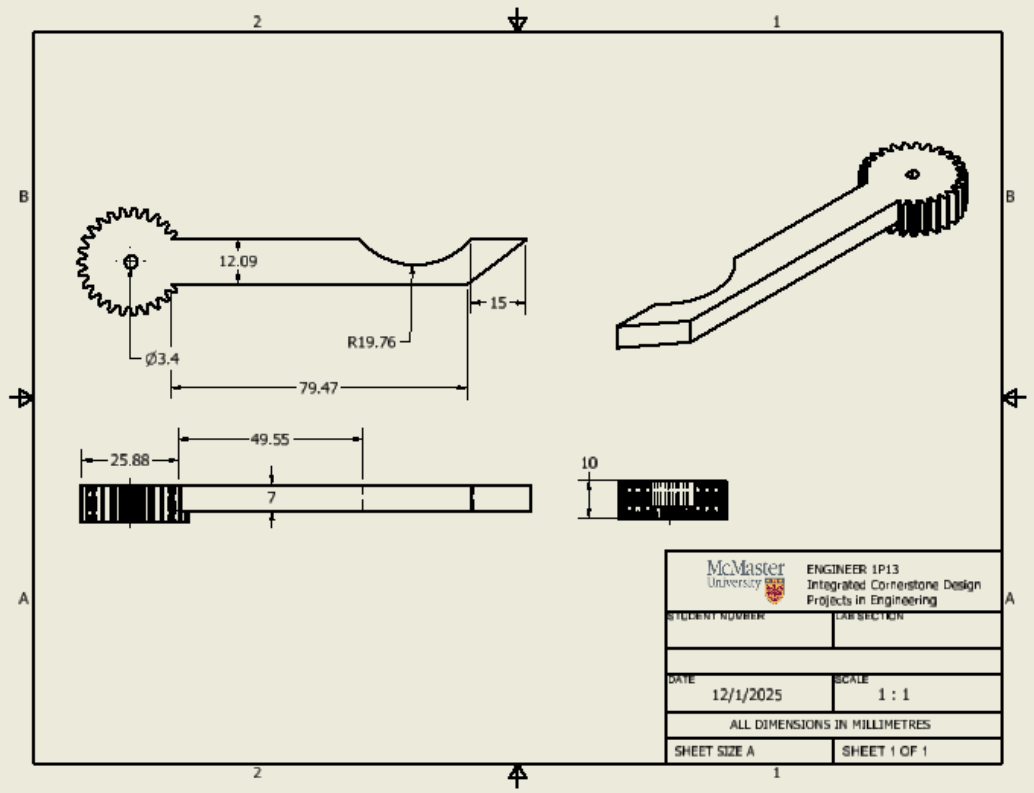


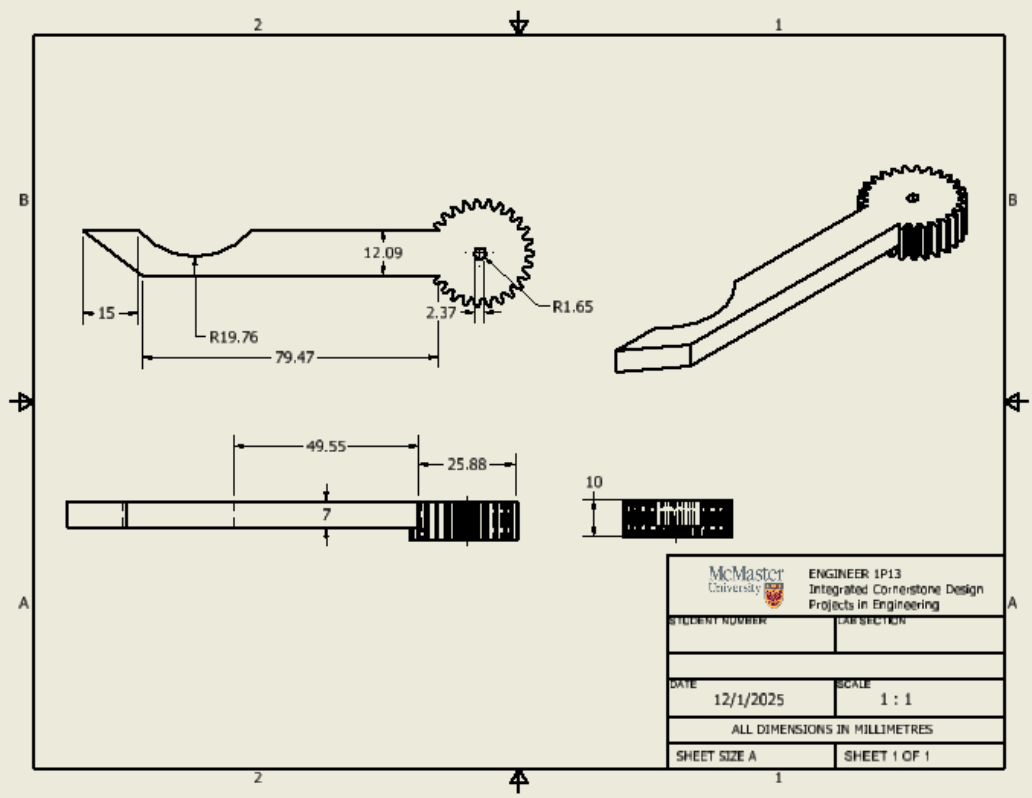




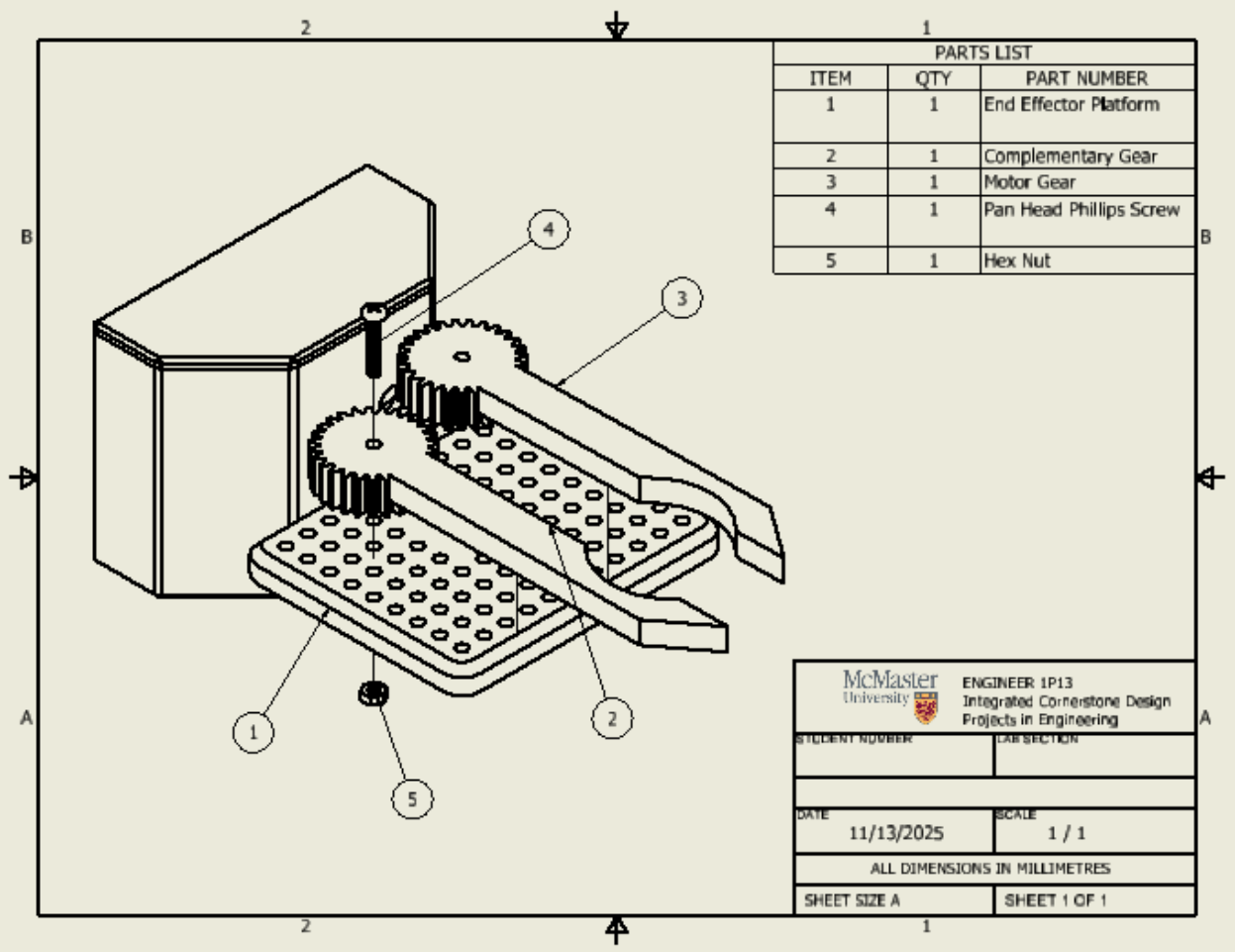


Fully dimensioned Engineering Drawings:





An assembly exploded view drawing:



Appendix E: Design Studio Worksheets

Team Milestones

MILESTONE 0 (TEAM): COVER PAGE

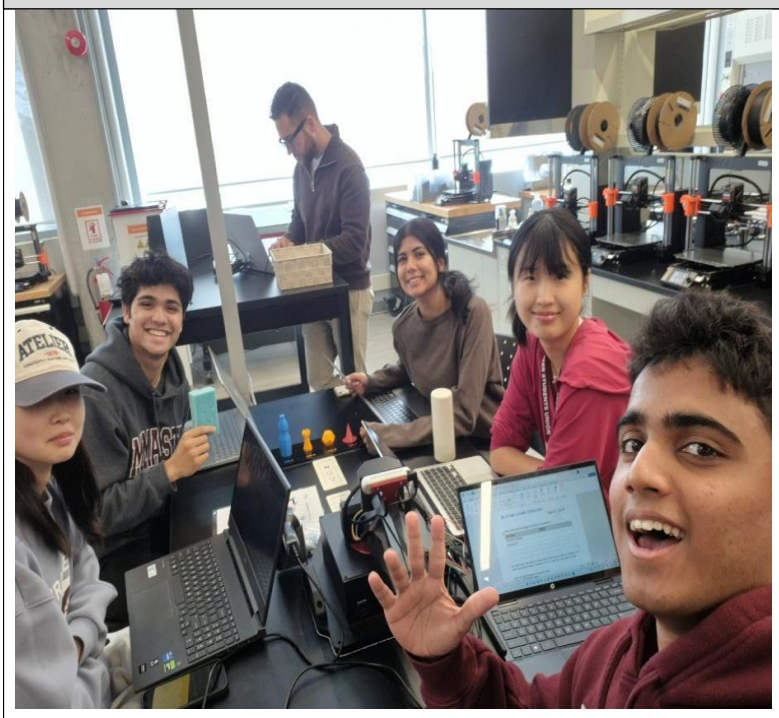
Team ID: Thur-04

Please list full names and MacIDs of all *present* Team Members

Full Name:	MacID:
Shivank Sunil Nambiar	nambis2
Ayla Nasir	nasira20
Raphael Serrao	serrar2
Wenyue Jiang	jiangw74
Sherry Zhao	zhaot75

Any student that is ***not*** present for Design Studio will not be given credit for completion of the worksheet and may be subject to a deduction to their P-1 grade. See course outline for details.

Insert your Team Portrait in the dialog box below

Team Portrait


MILESTONE 0 – TEAM CHARTER

Team ID: Thur-04

Project Leads:

Identify team member details (Name and MacID) in the space below.

Role:	Team Member Name:	MacID
Manager	Ayla Nasir	nasira20
Administrator	Wenyue Jiang	jiangw74
Coordinator	Sherry Zhao	zhaot75
Subject Matter Expert	Shivank Sunil Nambiar	nambis2
Subject Matter Expert	Raphael Serrao	serrar2

MILESTONE 0 – PRELIMINARY GANTT CHART (TEAM MANAGER ONLY)

Team ID: Thur-04

Only the **Project Manager** is completing this section!

Full Name of Team Manager:	MacID:
Ayla Nasir	nasira20

Preliminary Gantt Chart

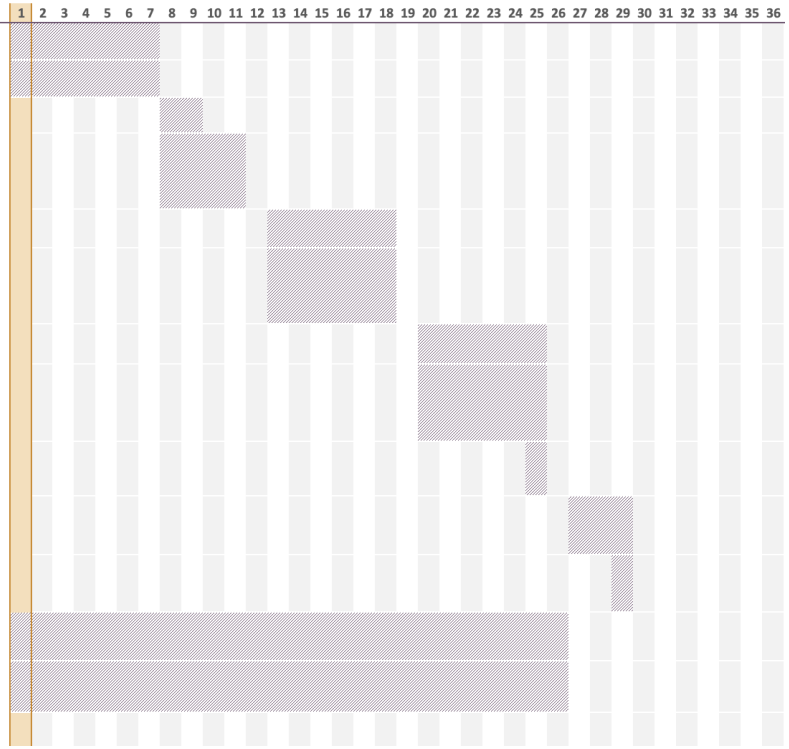
Project 1 Planner

Select a period to highlight at right. A legend describing the charting follows.

Period Highlight: 1

Plan Duration | Actual Start | % Complete | Actual (beyond plan) | % Complete (beyond plan)

ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Milestone 0	1	7	0	0	100%
Milestone 1	1	7	0	0	100%
Milestone 2	8	2	0	0	0%
Work Period / Fabrication Time	8	4	0	0	0%
Milestone 3	13	6	0	0	0%
Work Period / Informal TA Check-in	13	6	0	0	0%
Milestone 4	20	6	0	0	0%
Work Period / Informal TA Check-in	20	6	0	0	0%
Final Gantt Chart	25	1	0	0	0%
Project Demonstrations	27	3	0	0	0%
Peer evaluation	29	1	0	0	0%
Design Summary	1	26	0	0	0%
Logbook of Meetings	1	26	0	0	0%



MILESTONE 1 (TEAM): COVER PAGE

Team ID: Thur-04

Please list full names and MacIDs of all *present* Team Members

Full Name:	MacID:
Ayla Nasir	nasira20
Wenyue Jiang	jiangw74
Raphael Serrao	serrar2
Sherry Zhao	zhaot75
Shivank Sunil Nambiar	nambis2

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MILESTONE 1 (STAGE 1) – PROBLEM STATEMENT, FUNCTIONS, OBJECTIVES AND CONSTRAINTS

Team ID: Thur-04

Write out your initial problem statement in the box below.

Initial Problem Statement
The current robotic end effectors struggle to safely package and move products of varying sizes. A robotic end effector that can package and transport items with unusual shapes effectively without damage will be designed. The design should be easy to operate and cost effective.

List out your functions, objectives and constraints in the box below.

Functions	Objectives	Constraints
• Package and transport diverse package sizes • Process order data • Employee login/ sign up • Producing order receipts	• To package and transport items with unusual shapes effectively without the need for manual labor. • Built with sustainable materials • Lift packages of large mass • Should package with correct order	• Allocating materials properly • Less operational cost • Less material waste • No damage during process • Less manual labor • Must be factory and user friendly • Reduce physical strain and improve safety

MILESTONE 1 (STAGE 2) – TA CHECK-IN

During and after your TA check-in document their feedback in the box below.

Feedback
Tip: Overall: should have more formal writing, no personal pronouns, use verbs instead Function part: N/A Constraint: Combine with safety purposes Objective: Injury concerns more fits in constraint part

List out action items to address any of your team's concerns and your TA's feedback

Action Items
• Adjust overall vocabulary to fit professional environment • Keeping safety as a priority going forward while coming up with ideas

MILESTONE 2 (TEAM) – COVER PAGE

Team ID: Thur-04

Please list full names and MacIDs of all *present* Team Members

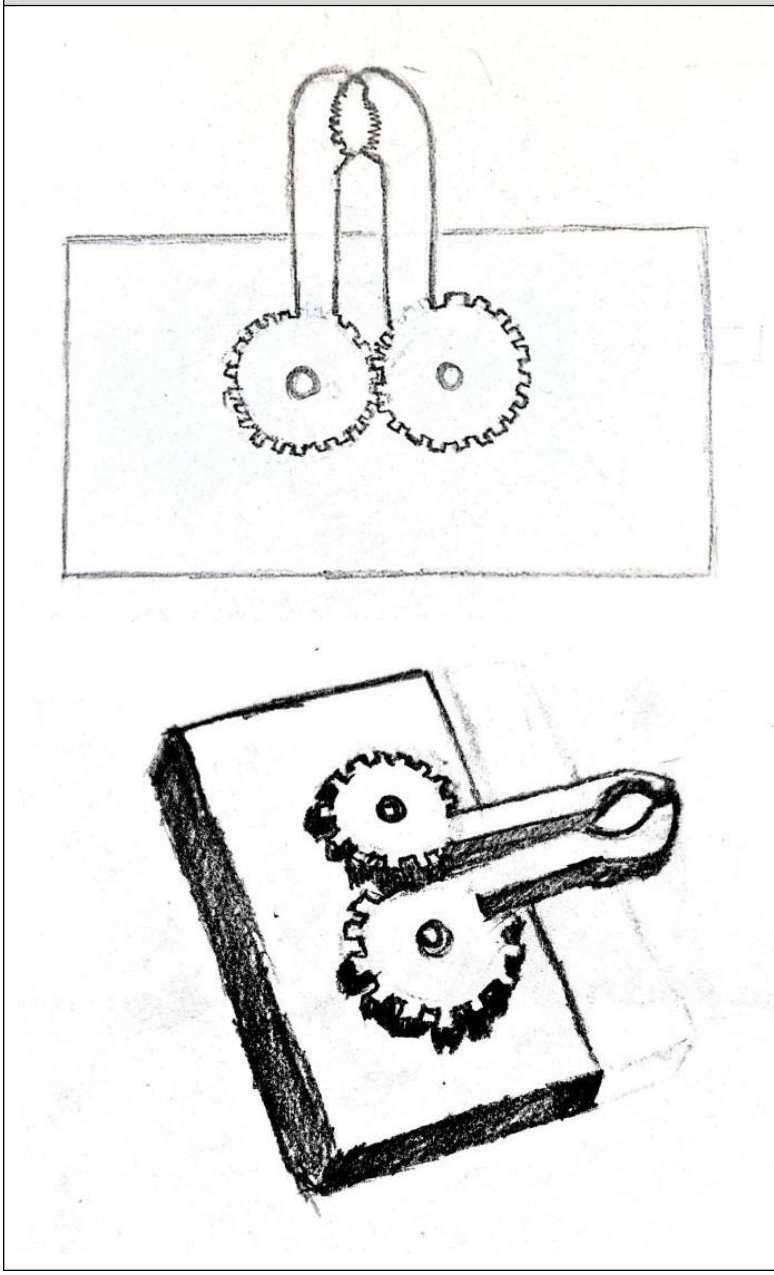
Full Name:	MacID:
Ayla Nasir	nasira20
Wenyue Jiang	Jiangw74
Sherry Zhao	zhaot75
Shivank Nambiar	nambis2
Raphael Serrao	serrar2

Any student that is ***not*** present for Design Studio will not be given credit for completion of the worksheet and may be subject to a deduction to their P-1 grade. See course outline for details.

MILESTONE 2 (STAGE 2) – MECHANISM DECISION AND CAD MODEL

Team ID: Thur-04

Attach your teams chosen finalized mechanism concept sketch in the following box

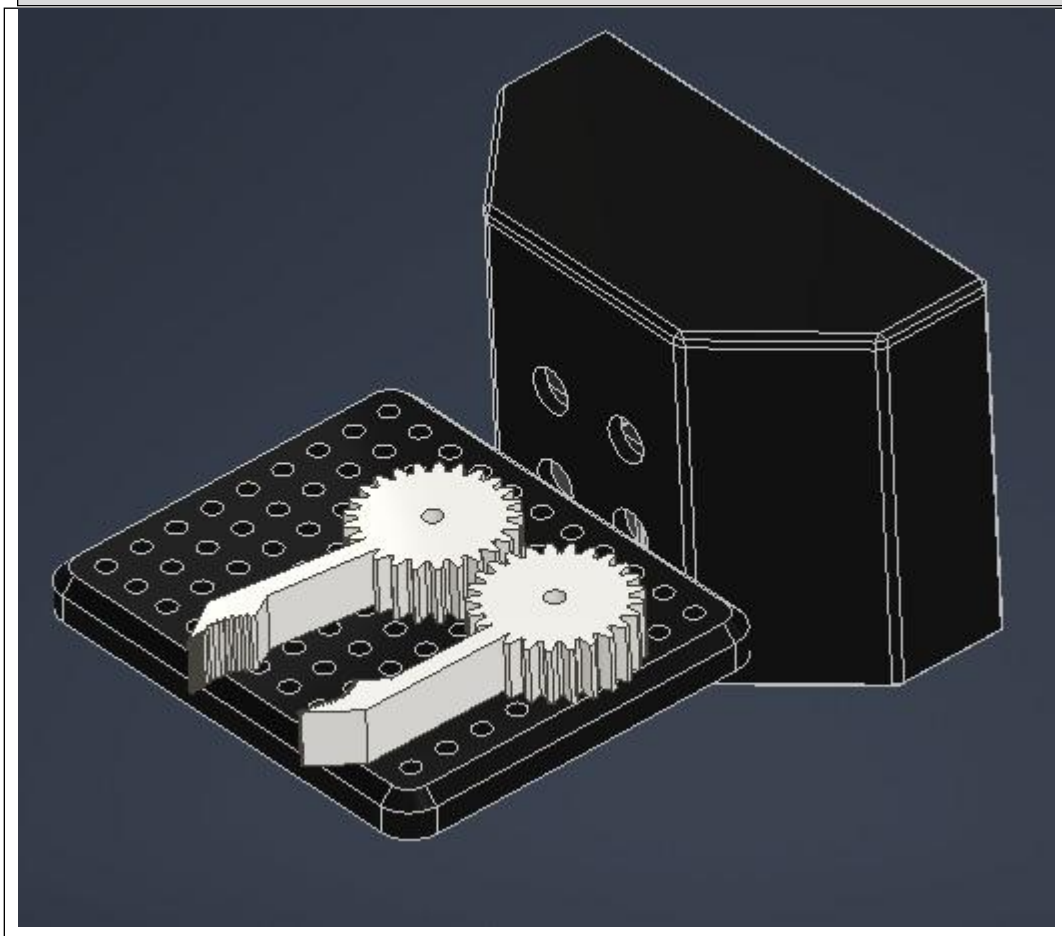
Finalized Concept Sketch


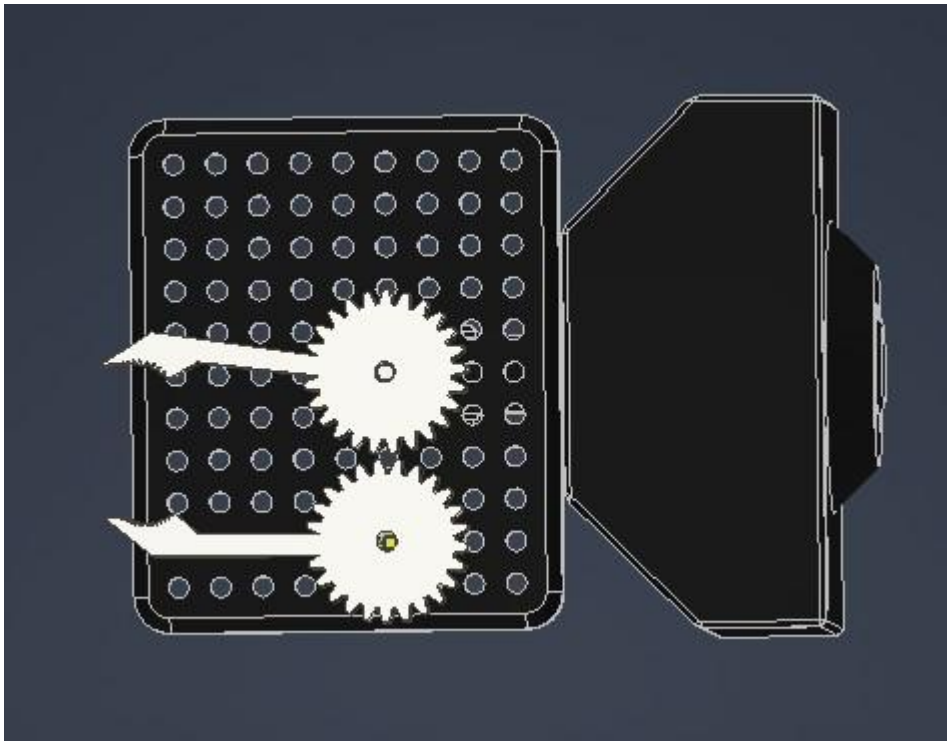
Justify why your team chose this design

Justification
Our team selected this claw gear-driven gripper design because it can effectively pick up objects of various shapes and sizes reliably. The two "jaws" are connected by gears, resulting in symmetrical and synchronous motion that automatically centers the object while gripping. This symmetry enables the gripper to adjust to a wide range of items while retaining a balanced force distribution. The curved jaw shape improves surface contact, increasing stability and lowering the chance of slippage when the object moves from one spot to another. The sharper point at the end of the "jaws" enables for more precise picking up of items, such as the cone. Furthermore, the simple mechanical system, which is powered by a single rotating mechanism and contains no extra components, provides for material resourcefulness, and a small margin for error.

Attach multiple screenshots of different views of your CAD model to the following box

CAD Model Screenshots





Select one or more objectives for your design and explain how you plan to further optimize it for each.

Justification
The design should be able to effectively grip and lift items of varying sizes and weights. To optimize it to reach this objective, a precision grip will be created at the end of the "jaws" by making sure they are narrow for the more pointy objects, and they will be wider near the gears such that it can still pick up the wider objects as well. Trial and error can be used to determine the optimal width of the jaws. Another goal for the design was to have it made of sustainable materials. To further optimize this, every part used in the design will have a specific purpose that it effectively meets while the Q-Arm is in use. If this isn't the case, the part will be altered or gotten rid of it entirely. Another objective is to ensure that the gripper operates with maximum efficiency. Two approaches will be implemented to achieve this goal. The current gear has 25 teeth; gears with fewer teeth require greater force to rotate and are more susceptible to damage. To enhance efficiency, gears with a higher tooth count will be tested. Increasing the number of teeth distributes the load more evenly and enables smoother rotation, though it also requires additional material. These trade-offs will be evaluated to determine the most efficient configuration for the gripper mechanism. The second approach focuses on minimizing friction. Since the end effector will be 3D printed, the gear surfaces are expected to be relatively rough, which can increase friction and energy loss. To address this, the gear teeth will be sanded and smoothed after printing to reduce surface roughness.

WORK PERIOD: FABRICATION TIME – COVER PAGE

Team ID: Thur-04

Please list full names and MacIDs of all *present* Team Members

Full Name:	MacID:
Ayla Nasir	nasira20
Wenyue Jiang	Jiangw74
Sherry Zhao	zhaot75
Shivank Nambiar	nambis2
Raphael Serrao	serrar2

Any student that is ***not*** present for Design Studio will not be given credit for completion of the worksheet and may be subject to a deduction to their P-1 grade. See course outline for details.

WORK PERIOD: FABRICATION TIME (STAGE 3) –

TA CHECK-IN

Team ID: Thur-04

During and after your TA check-in document their feedback in the box below.

Feedback
• The sketch was in the iam file, but it should be in the ipt file • Spur gear should be in ipt file • Enlarge the 2 parts in the gear feature because the 2 gears are not connecting and working with one another to move • Extend the arms so they can properly pick up items, right now the arms are too thin and small • Use calliper for accurate measurements to increase reliability of parts to be the proper size

List out action items to address any of your team's concerns and your TA's feedback

Action Items
• Measuring tools in inventor will be used to ensure parts work in harmony and meet their role in the mechanism • The arms will be extended and thicken so they can properly pick up the items • The shape of the claws will be changed from curve spurs to flatter spurs to be more reliable and versatile • The area of the spurs around the claws will also be increased in order to pick up the level 4 design

MILESTONE 3 (TEAM) – COVER PAGE

Team ID: Thur-04

Please list full names and MacIDs of all *present* Team Members

Full Name:	MacID:
Ayla Nasir	nasira20
Wenyue Jiang	Jiangw74
Sherry Zhao	zhaot75
Shivank Nambiar	nambis2
Raphael Serrao	serrar2

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MILESTONE 3 (STAGE 1) – FUNCTION ASSIGNMENT

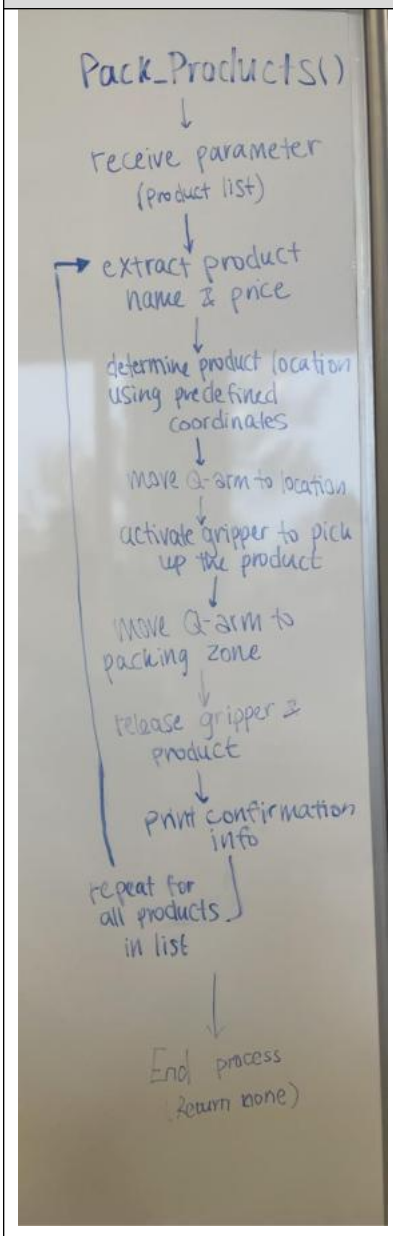
Team ID: Thur-04

Using the table below assign each function to one member of your team.

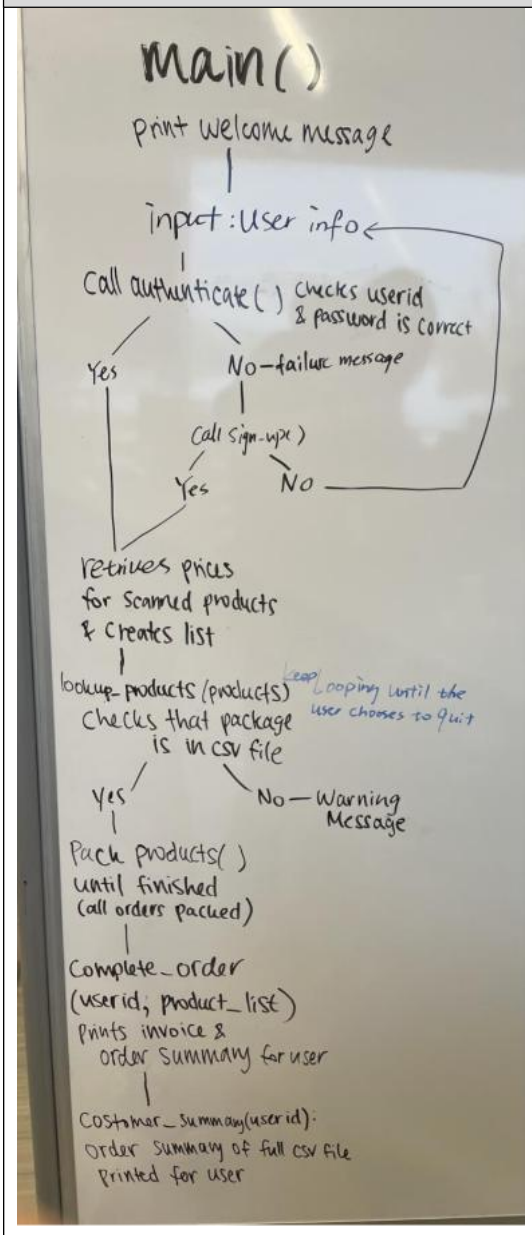
Function	Full Name	MacID
sign_up()	Ayla Nasir	nasira20
authenticate()	Shivank Nambiar	nambis2
lookup_products()	Raphael Serrao	serrar2
complete_order()	Sherry Zhao	zhaot75
customer_summary()	Wenyue Jiang	Jiangw74

MILESTONE 3 (STAGE 2) – FLOWCHART

Insert images of flowcharts in the boxes below for the two group functions: pack_products() and main().

pack_products()


main()


main()
print welcome message
input: user info
Call authenticate( ) checks userid & password is correct
Yes
No – failure message
Call sign_up( )
Yes
No
retrives prices for scanned products & creates list
lookup_products (products) Looping until the user chooses to quit
Checks that package is in csv file
Yes
No – Warning Message
Pack products( ) until finished (all orders packed)
Complete_order (userid, product_list)
Prints invoice & order summary for user
Costomer_summary(userid):
Order summary of full csv file Printed for user

MILESTONE 3 (STAGE 3) –TA CHECK-IN

During and after your TA check-in document their feedback in the box below.

Feedback
• The flowchart should include the standard flowchart symbols • In the future, a software could be used to make flowchart look more professional

List out action items to address any of your team's concerns and your TA's feedback

Action Items
• In the future, a flowchart software will be used to ensure the flowchart has the correct symbols, and looks professional

WORK PERIOD: CODE YOUR FUNCTION – COVER PAGE

Team ID: Thur-04

Please list full names and MacIDs of all *present* Team Members

Full Name:	MacID:
Ayla Nasir	nasira20
Wenyue Jiang	Jiangw74
Sherry Zhao	zhaot75
Shivank Nambiar	nambis2
Raphael Serrao	serrar2

Any student that is ***not*** present for Design Studio will not be given credit for completion of the worksheet and may be subject to a deduction to their P-1 grade. See course outline for details.

WORK PERIOD: CODE YOUR FUNCTION (STAGE 1) –

TA CHECK-IN

Team ID: Thur-04

During and after your TA check-in document their feedback in the box below.

Feedback
• Ensure the Q-Arm code for any movement is properly implemented into the pack_products(): function • Main(): function may have to be refined as we implement the rest of our individual code

List out action items to address any of your team's concerns and your TA's feedback

Action Items
• The proper Q-Arm code for the movement required will be written out in the code • Main(): function will be revisited as more code is added to ensure all code

MILESTONE 4 (TEAM) – COVER PAGE

Team ID: Thur-04

Please list full names and MacIDs of all *present* Team Members

Full Name:	MacID:
Ayla Nasir	nasira20
Wenyue Jiang	Jiangw74
Sherry Zhao	zhaot75
Shivank Nambiar	nambis2
Raphael Serrao	serrar2

Any student that is ***not*** present for Design Studio will not be given credit for completion of the worksheet and may be subject to a deduction to their P-1 grade. See course outline for details.

MILESTONE 4 (STAGE 1) – Q-ARM CODE

Insert screenshots of your Q-Arm code to the box below.

Q-Arm Code Screensho

```
#Sponge code
arm.home()
sleep(1)
arm.set_arm_position([0.5863166758552483, 0.17349144868212818, 0.12230771334418886])
sleep(1)
arm.rotate_gripper(60)
sleep(1)
sleep(1)
arm.get_arm_position()
sleep(1)
'''
arm.rotate_gripper(90)
sleep(1)
arm.rotate_elbow(-30)
sleep(1)
arm.home()
arm.home()
sleep(1)
arm.rotate_base(17)
sleep(1)
arm.rotate_shoulder(30)
sleep(1)
arm.rotate_shoulder(10)
sleep(1)
arm.rotate_gripper(40)
sleep(1)
arm.rotate_elbow(-10)
sleep(1)
arm.rotate_shoulder(2.5)
sleep(1)
arm.get_arm_position()

arm.rotate_shoulder(2.5)
sleep(1)
arm.rotate_elbow()
sleep(1)
sleep(1)
arm.rotate_gripper(95)
sleep(1)
arm.rotate_elbow(-10)
sleep(1)
arm.rotate_shoulder(-20)
'''
# Barcode code: scan_barcode()
'''scan_barcode()
Click inside the shell window and scan the barcode: Sponge Bottle Labubu Bowl
'Sponge Bottle Labubu Bowl '''
# ----------------------------------------------------------------------------
```

```
# ----------------------------------------------------------------------------
#Bottle Code:
'''
arm.home()
sleep(1)
arm.set_arm_position([0.5954305781626856, 0.1151475142434109, 0.11145655737948598])
sleep(1)
arm.rotate_gripper(90)

arm.rotate_base(11)
sleep(1)
arm.rotate_shoulder(30)
sleep(1)
arm.rotate_shoulder(10)
sleep(1)
arm.rotate_elbow(-10)
sleep(1)
arm.rotate_shoulder(5)
sleep(1)
arm.rotate_elbow(-10)
sleep(1)
arm.get_arm_position()
'''
```

MILESTONE 4 (STAGE 3) – TA CHECK-IN

During and after your TA check-in document their feedback in the box below.

Feedback
- The positioning of the Q-Arm (where it moves) needs to be more accurate and efficient - The final code should follow a logical order of when the functions are called (eg. sign_up(): function should be around the top of the program code, whereas main(): would be near the end) - The scan_barcode() code provided needs to be implemented in the python shell, and should be able to retrieve the name of the products

List out action items to address any of your team's concerns and your TA's feedback

Action Items
- The code for more precise and correct positioning of the Q-Arm to relocate each object will be properly implemented - When the individual codes and any other required code are placed together, it will follow a logical flow - The scan_barcode() function will be implemented and tested to ensure that it works correctly.

Ayla Nasir

MILESTONE 1 (INDIVIDUAL) – COVER PAGE

Team ID: *Thur-04*

Please list full name and MacID.

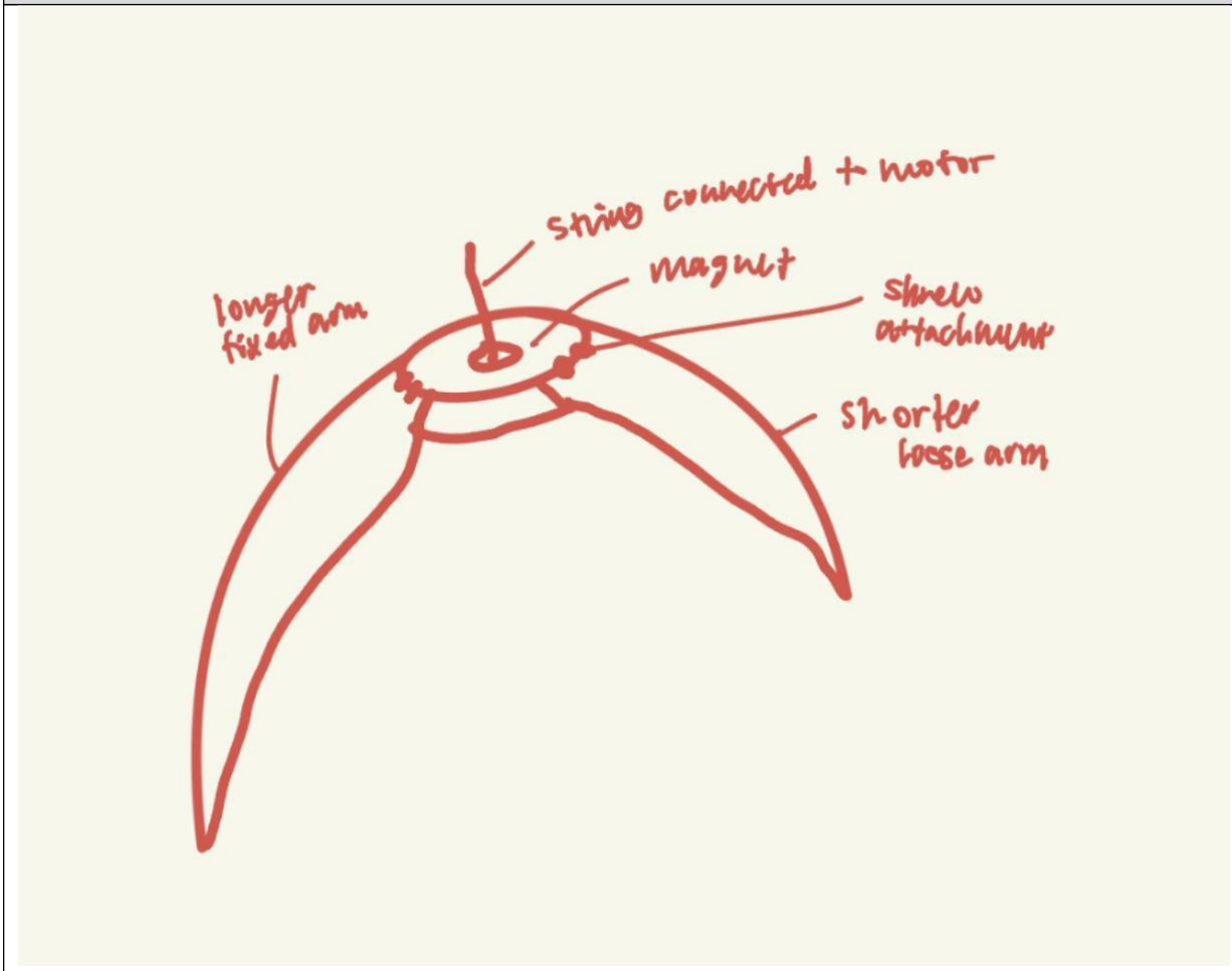
Full Name:	**MacID:**
Ayla Nasir	*nasira20*

MILESTONE 1 (STAGE 2) – MECHANISM CONCEPT SKETCHES

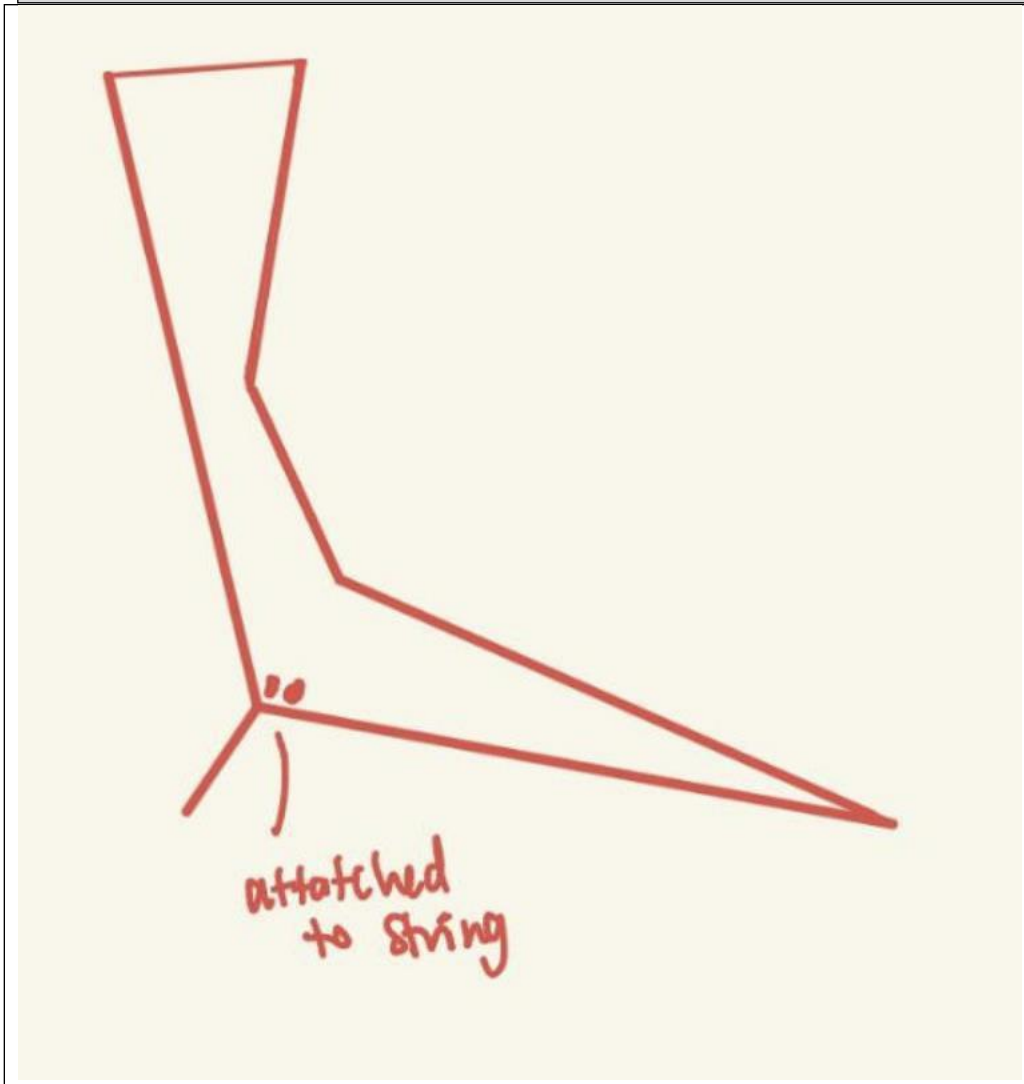
Team ID: Thur-04

Insert images of your initial mechanism sketches in the box below.

Sketch #1



Sketch #2



Select one or more objectives for your design and explain how you plan to optimize it for each.

Justification
One objective for my design is for the attachment to be versatile in its ability to pick up objects of varying shapes and sizes. Both sketches were created with flexibility in mind, ensuring they can adapt to different forms. Another objective is to use a sustainable amount of material. To achieve this, I avoided adding unnecessary parts with no clear function so that every component in my designs serves a specific purpose.

MILESTONE 2 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

Full Name:	**MacID:**
Ayla Nasir	*nasira20*

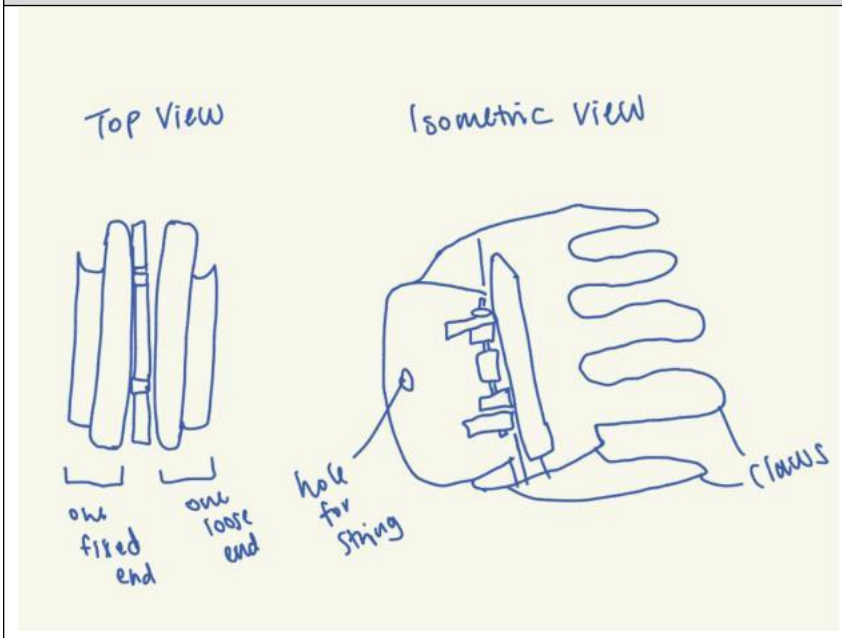
MILESTONE 2 (STAGE 1) – FINALIZED CONCEPT SKETCHES AND CAD MODEL WORKSHEET

Team ID: Thur-04

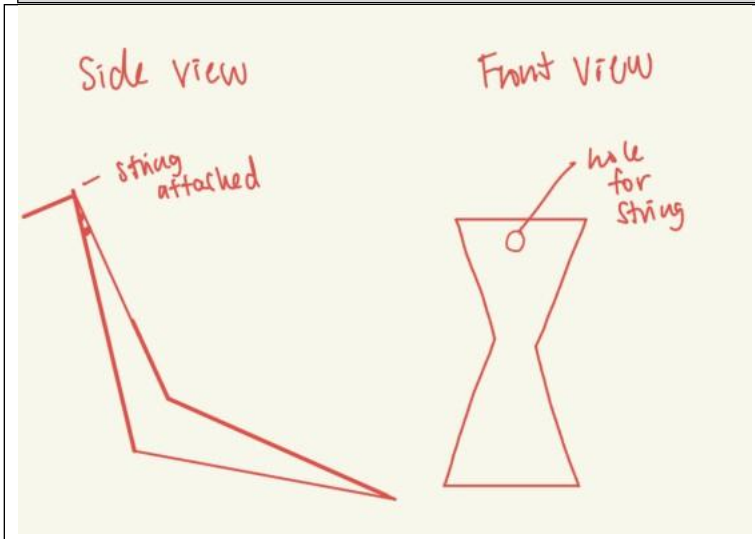
This is an individual deliverable and should be completed by each team member **prior** to Design Studio.

Attach images to the following three boxes of your mechanism sketch concepts and ideas.

Refined Sketch #1

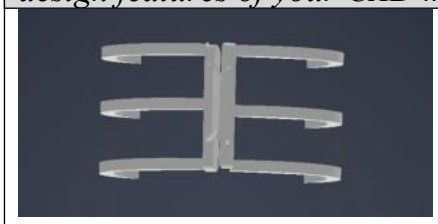


Refined Sketch #2



CAD Model Screenshots

Include screenshots of the top, front, right and isometric views of your CAD model. All design features of your CAD model should be visible.









Select one or more objectives for your design and explain how you plan to further optimize it for each.

Justification
One key objective for my design is to optimize the locking mechanism that engages after the claw wraps around an object. For the first sketch, which matches the CAD model, I plan to improve this by adding a locking feature, possibly using magnets between the two sides of the claw to create a more 8 secure grip on the object. For the second design, I want to refine the shape of the scoop and consider adding an extra component that helps objects slide in more easily and stay in place while being lifted and transferred to another platform.

MILESTONE 3 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

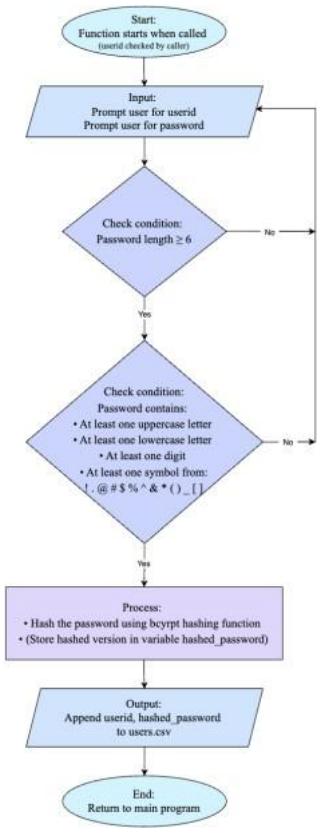
Full Name:	**MacID:**
Ayla Nasir	*nasira20*

MILESTONE 3 (STAGE 2) – FLOWCHART

Team ID: Thur-04

Insert an image of flowchart in the box for your assigned function. Also change the name of the function to match your assigned function.

sign_up()



Wenyue Jiang

MILESTONE 1 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

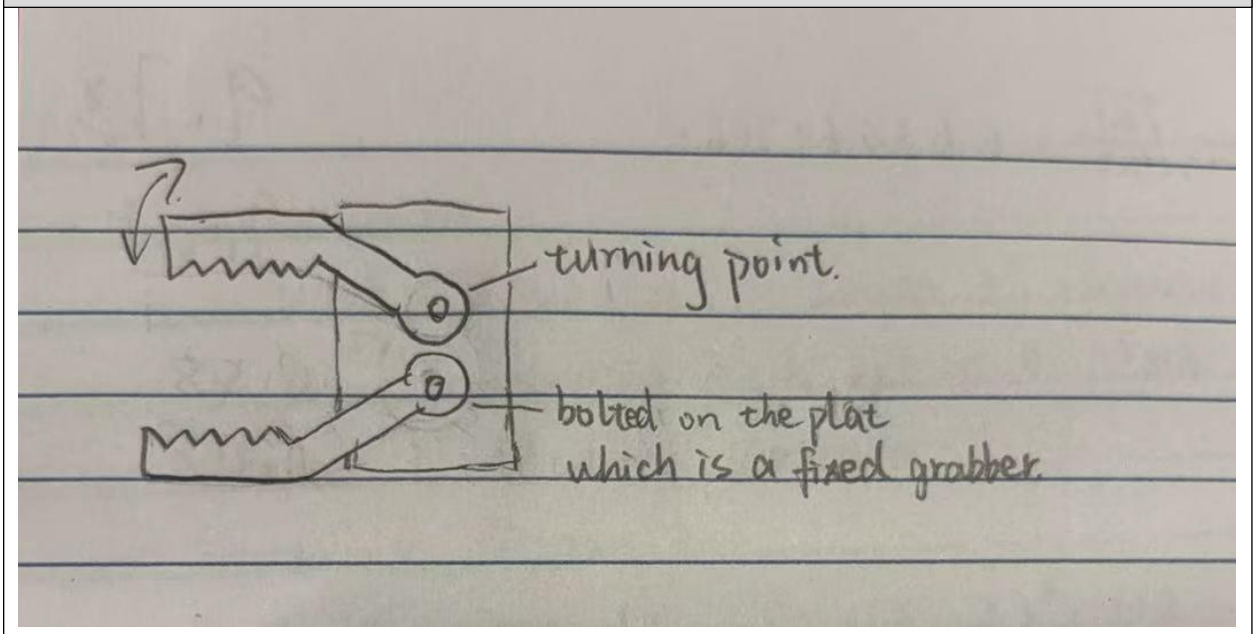
Full Name:	MacID:
Wenyue Jiang	Jiangw74

MILESTONE 1 (STAGE 2) – MECHANISM CONCEPT SKETCHES

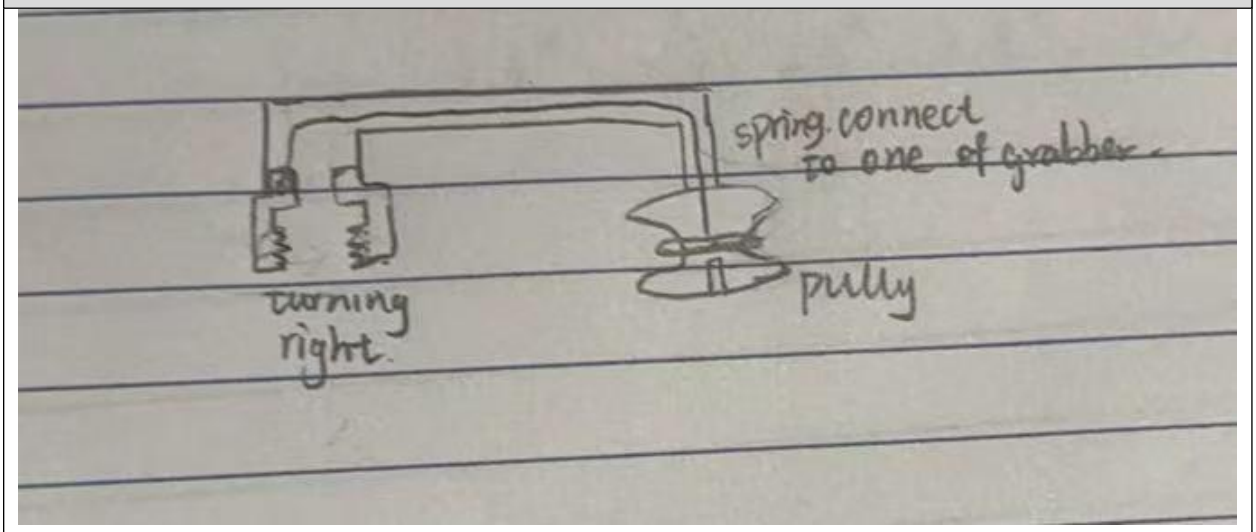
Team ID: Thur-04

Insert images of your initial mechanism sketches in the box below.

Sketch #1



Sketch #2



Select one or more objectives for your design and explain how you plan to optimize it for each.

Justification
Based on my stretch 1, the end of the grabber is designed as a sawtooth. - If it cannot grab the items with round surfaces (level 5 and 6), I will change the shape of the end as a circle. - If the grabber is not long enough to reach the items (since the Q-arm only can turn with limited angle, I am not sure how long my grabber should be), I will change the length of it or change the shape of it as the whole grabber extend outside the plat more and goes downward a little. Based on my stretch 2, the end of the grabber is also designed as a sawtooth. - If needed (same as the first stretch), I will change the shape of the end as a circle. - If the grabber can not reach the items (same reason as first stretch), I will change it to go downward instead upward or make the grabber longer.

MILESTONE 2 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

Full Name:	MacID:
Wenyue Jiang	Jiangw74

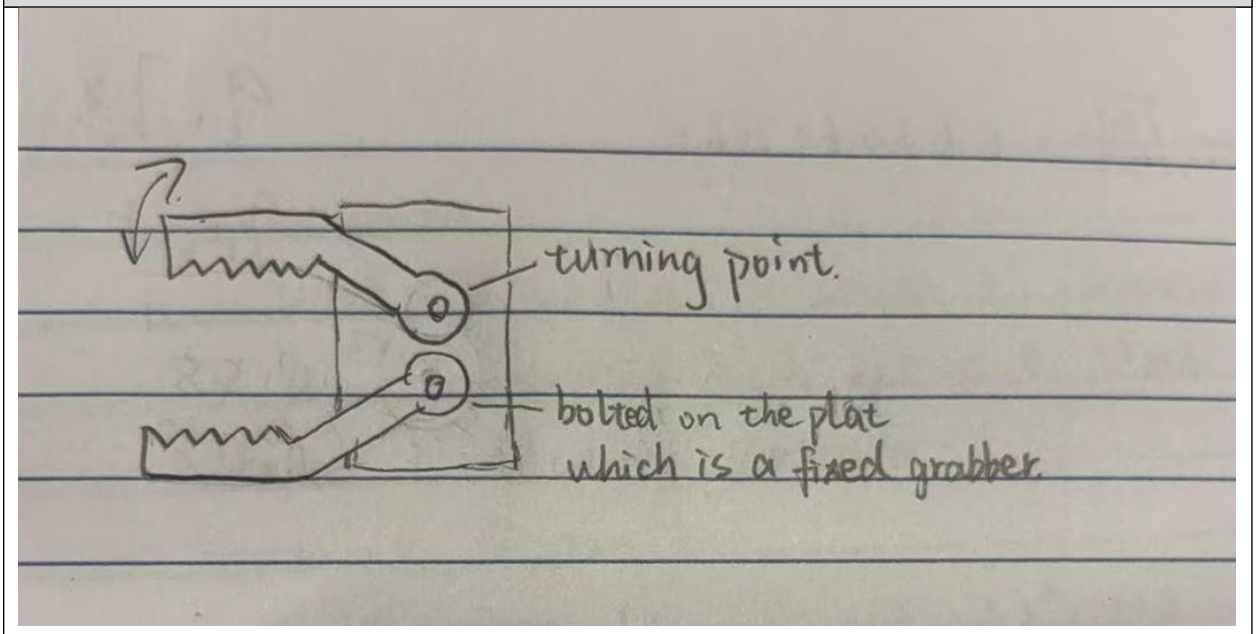
MILESTONE 2 (STAGE 1) – FINALIZED CONCEPT SKETCHES AND CAD MODEL WORKSHEET

Team ID: Thur-04

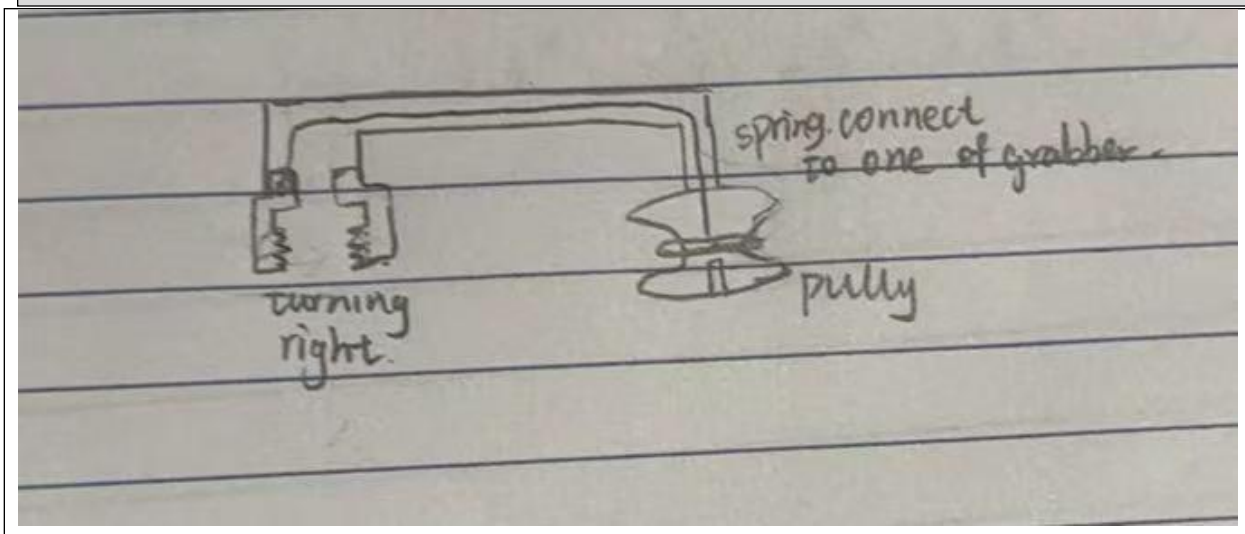
This is an individual deliverable and should be completed by each team member **prior** to Design Studio.

Attach images to the following three boxes of your mechanism sketch concepts and ideas.

Refined Sketch #1

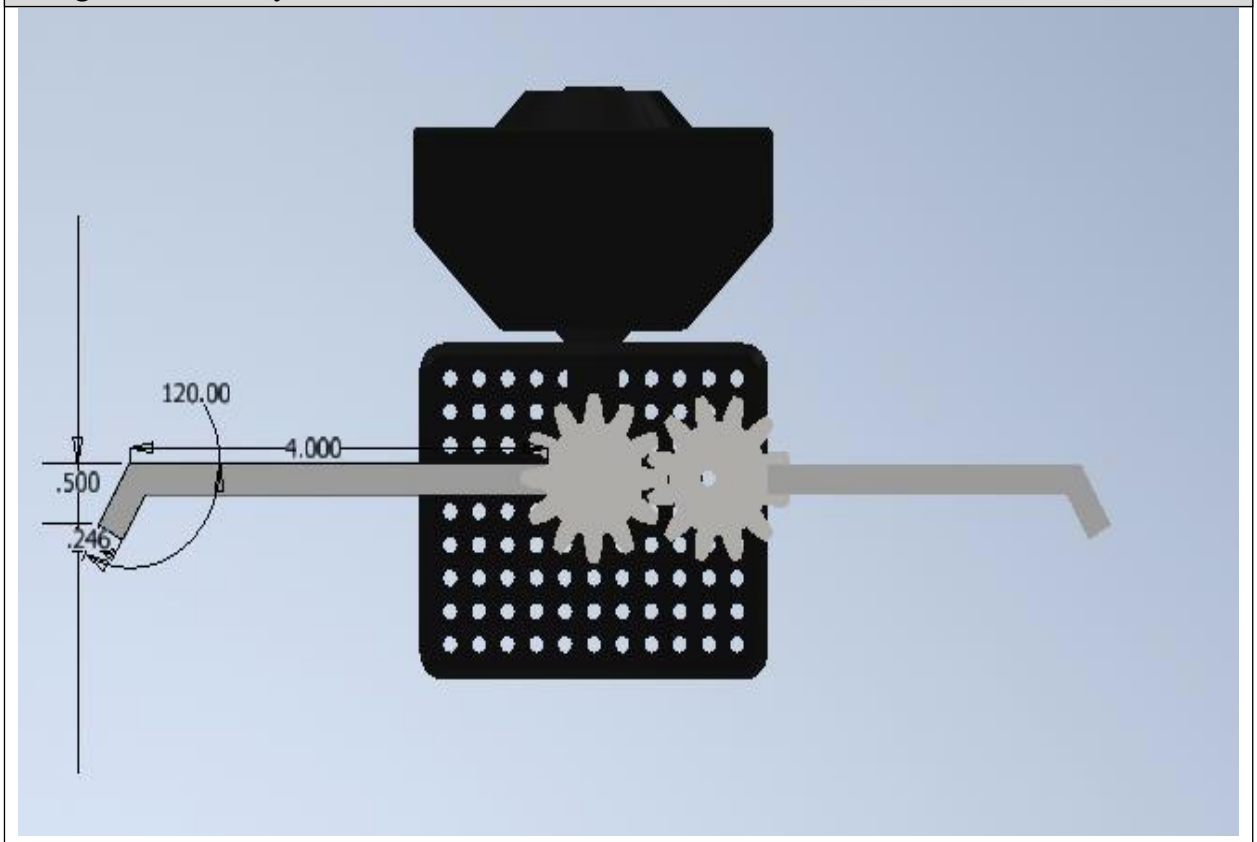


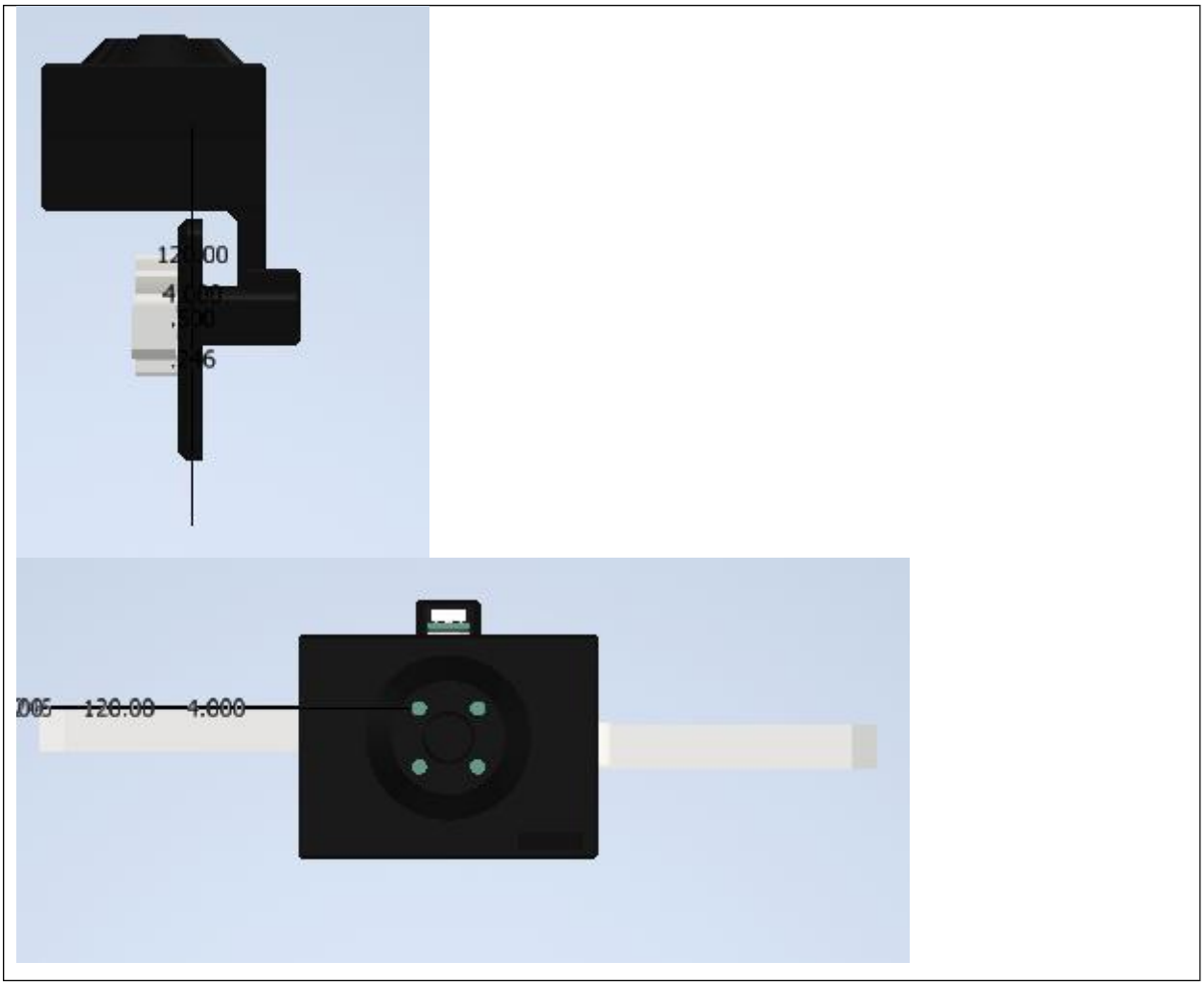
Refined Sketch #2

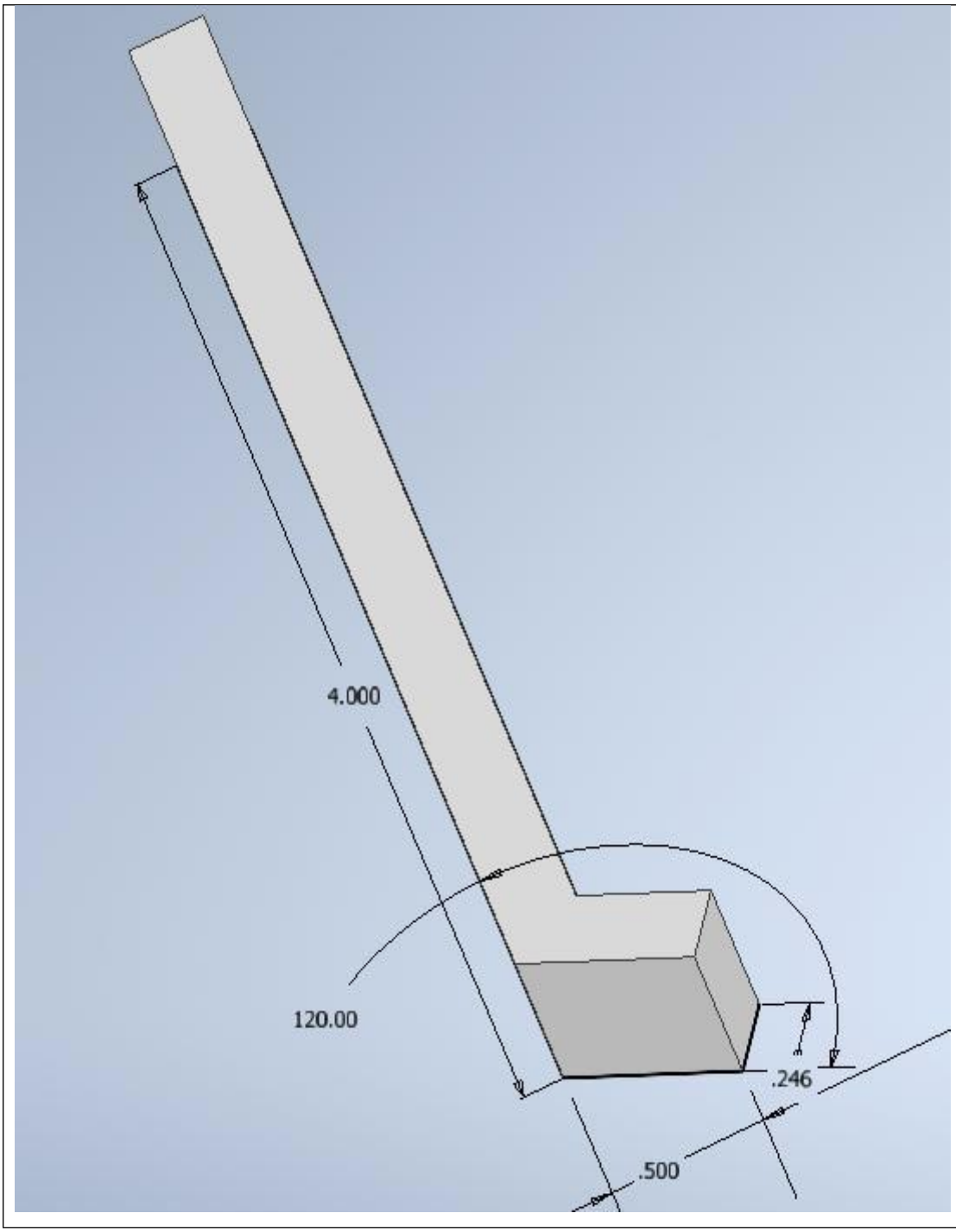


CAD Model Screenshots

Include screenshots of the top, front, right and isometric views of your CAD model. All design features of your CAD model should be visible.




120.00
4.000


4.000
120.00
.246
.500

Select one or more objectives for your design and explain how you plan to further optimize it for each.

Justification
Fully secure the objects during transport – add some teeth at the end of my grabber. Make the grabber more safe – make the grabber wider to ensure it would not broken when it is moving.

MILESTONE 3 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

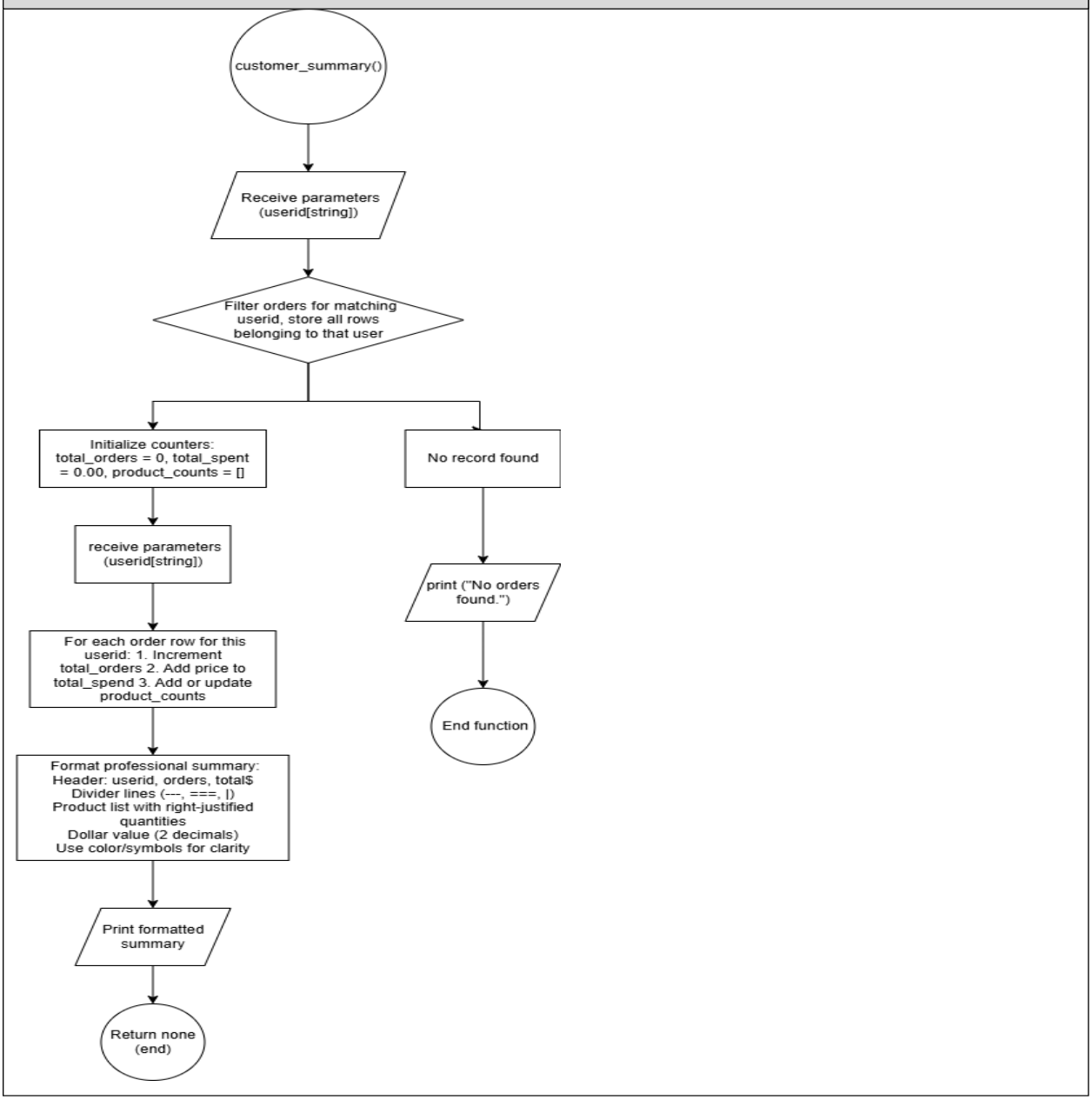
Full Name:	MacID:
Wenyue Jiang	Jiangw74

MILESTONE 3 (STAGE 2) – FLOWCHART

Team ID: Thur-04

Insert an image of flowchart in the box for your assigned function. Also change the name of the function to match your assigned function.

function_name() [customer_summary()]



Sherry Zhao

MILESTONE 1 (INDIVIDUAL) – COVER PAGE

Team ID: Thurs-04

Please list full name and MacID.

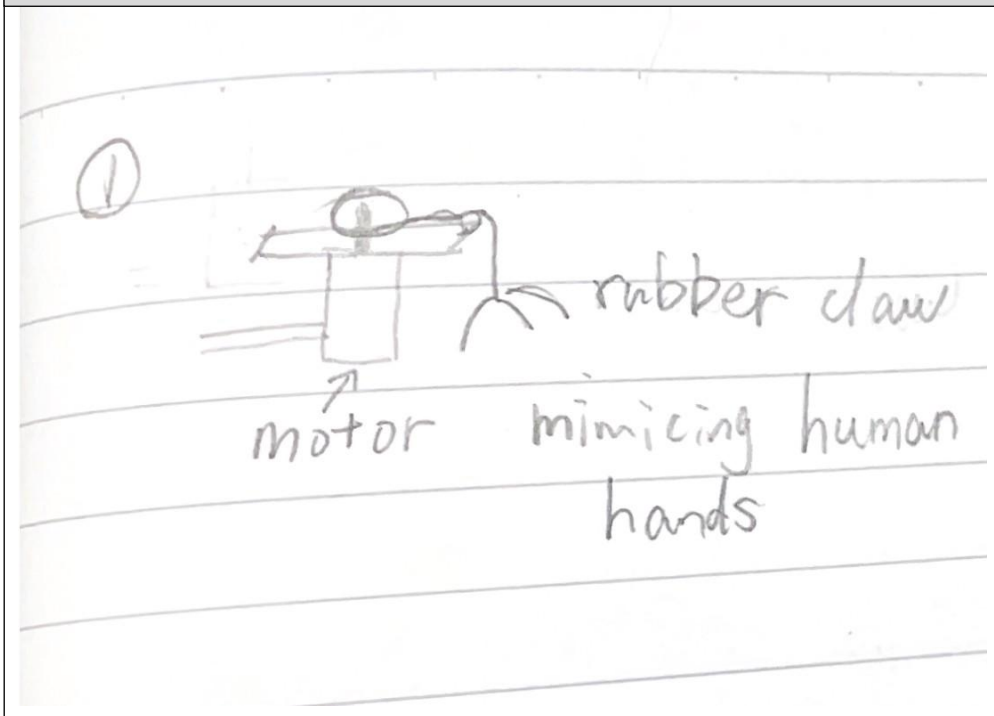
Full Name:	MacID:
Sherry Zhao	Zhaot75

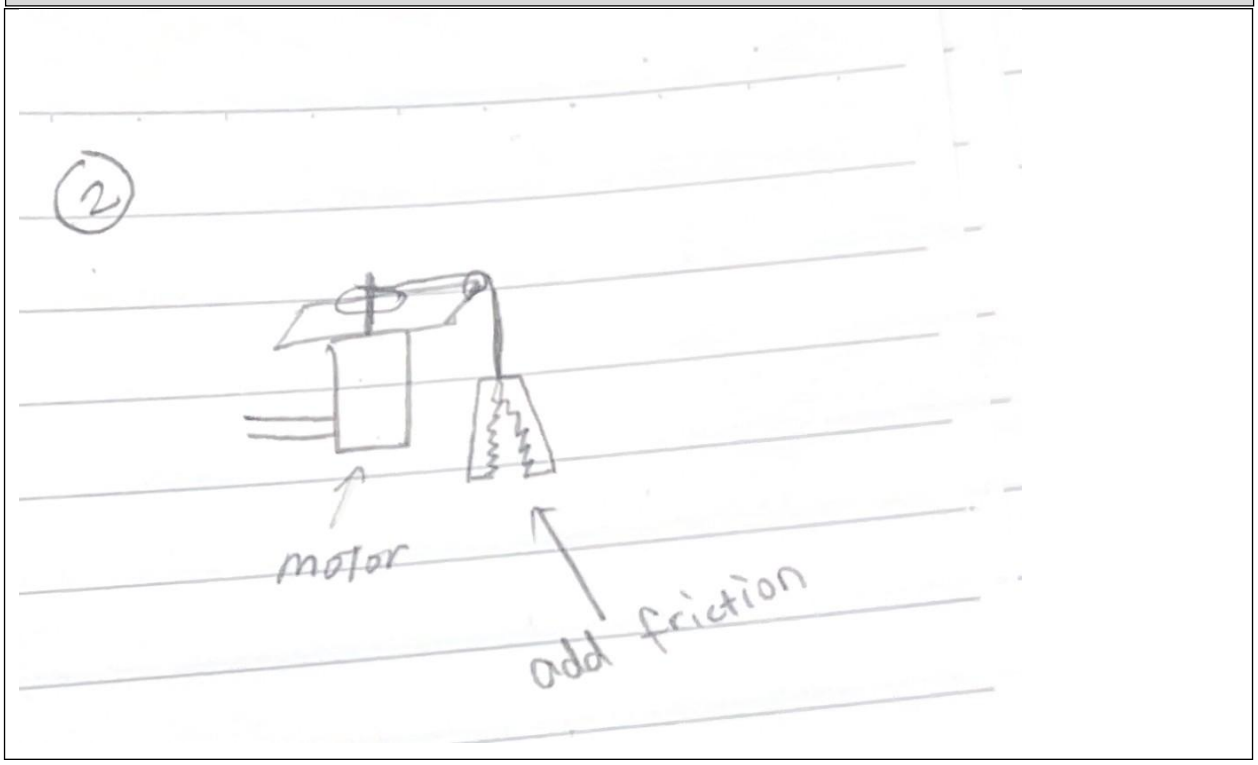
MILESTONE 1 (STAGE 2) – MECHANISM CONCEPT SKETCHES

Team ID: Thurs-04

Insert images of your initial mechanism sketches in the box below.

Sketch #1



Sketch #2


Select one or more objectives for your design and explain how you plan to optimize it for each.

Justification
For the design 2, to change the direction of the motor, putting 2 pulleys on top of the motor's plane would be a good idea, it could be attached by belt (or some hard to break materials). Then use a jagged claw with a Bluetooth attach to it to remotely control them,

MILESTONE 2 (INDIVIDUAL) – COVER PAGE

Team ID: Thurs-04

Please list full name and MacID.

Full Name:	MacID:
Sherry Zhao	Zhaot75

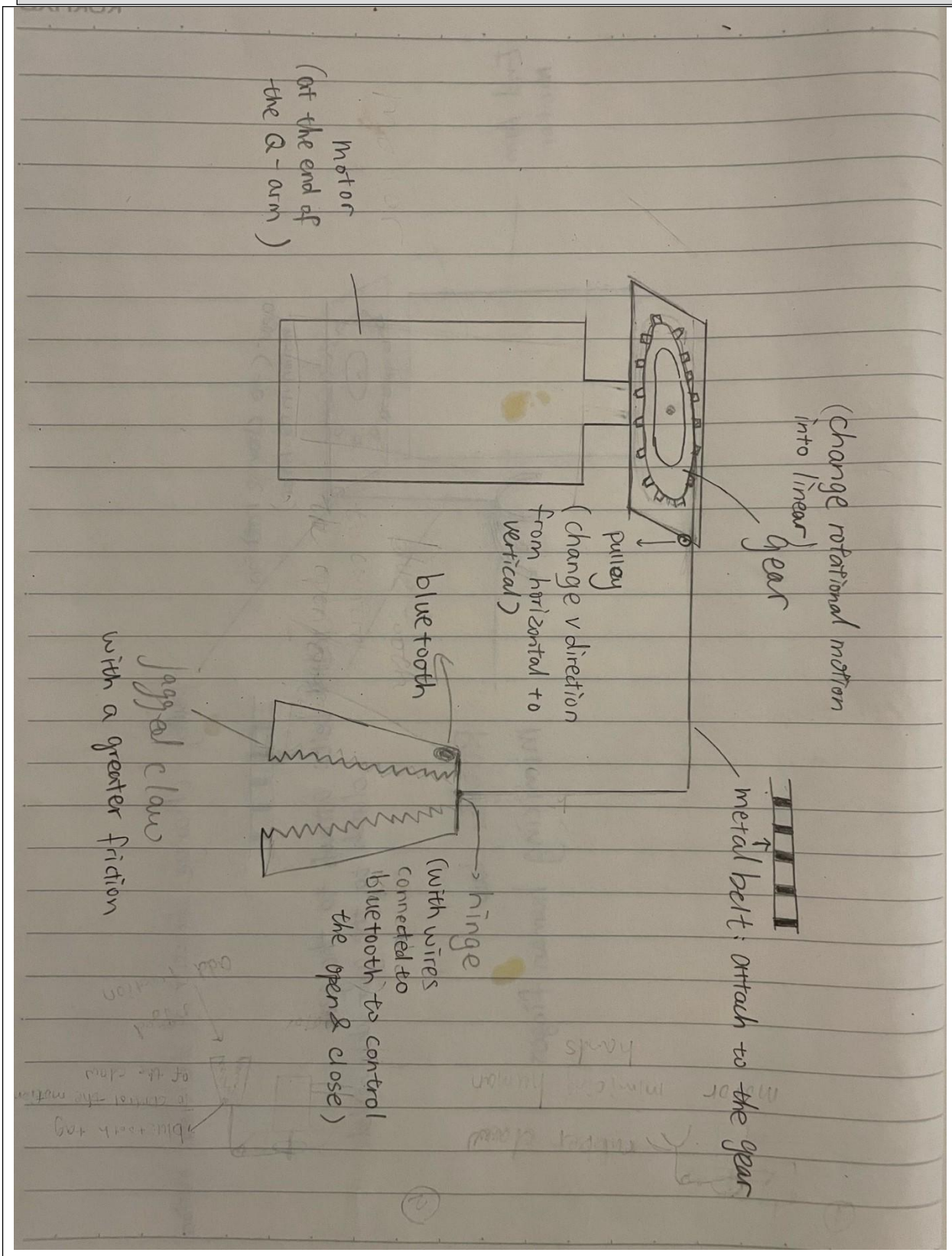
MILESTONE 2 (STAGE 1) – FINALIZED CONCEPT SKETCHES AND CAD MODEL WORKSHEET

Team ID: Thurs-04

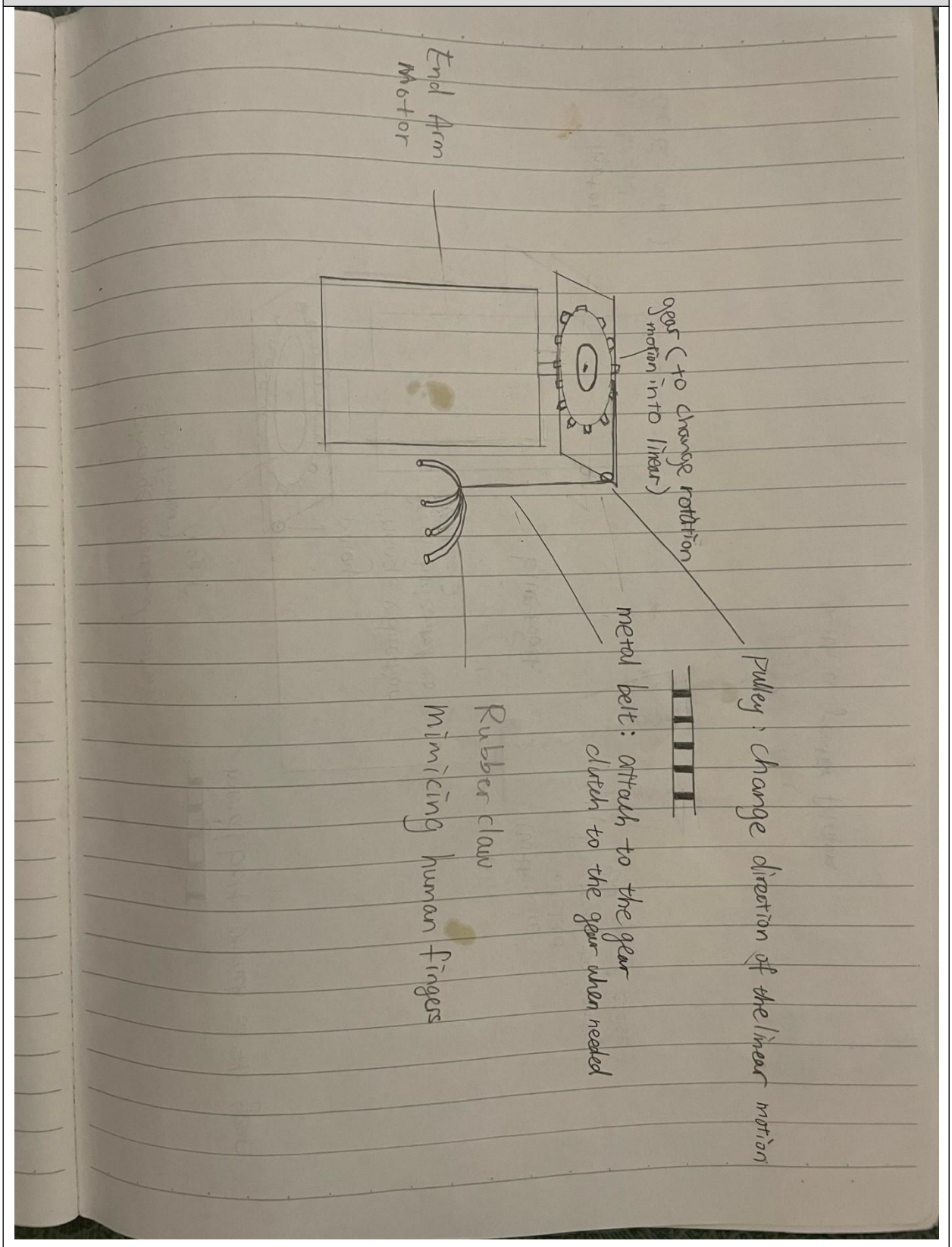
This is an individual deliverable and should be completed by each team member **prior** to Design Studio.

Attach images to the following three boxes of your mechanism sketch concepts and ideas.

Refined Sketch #1

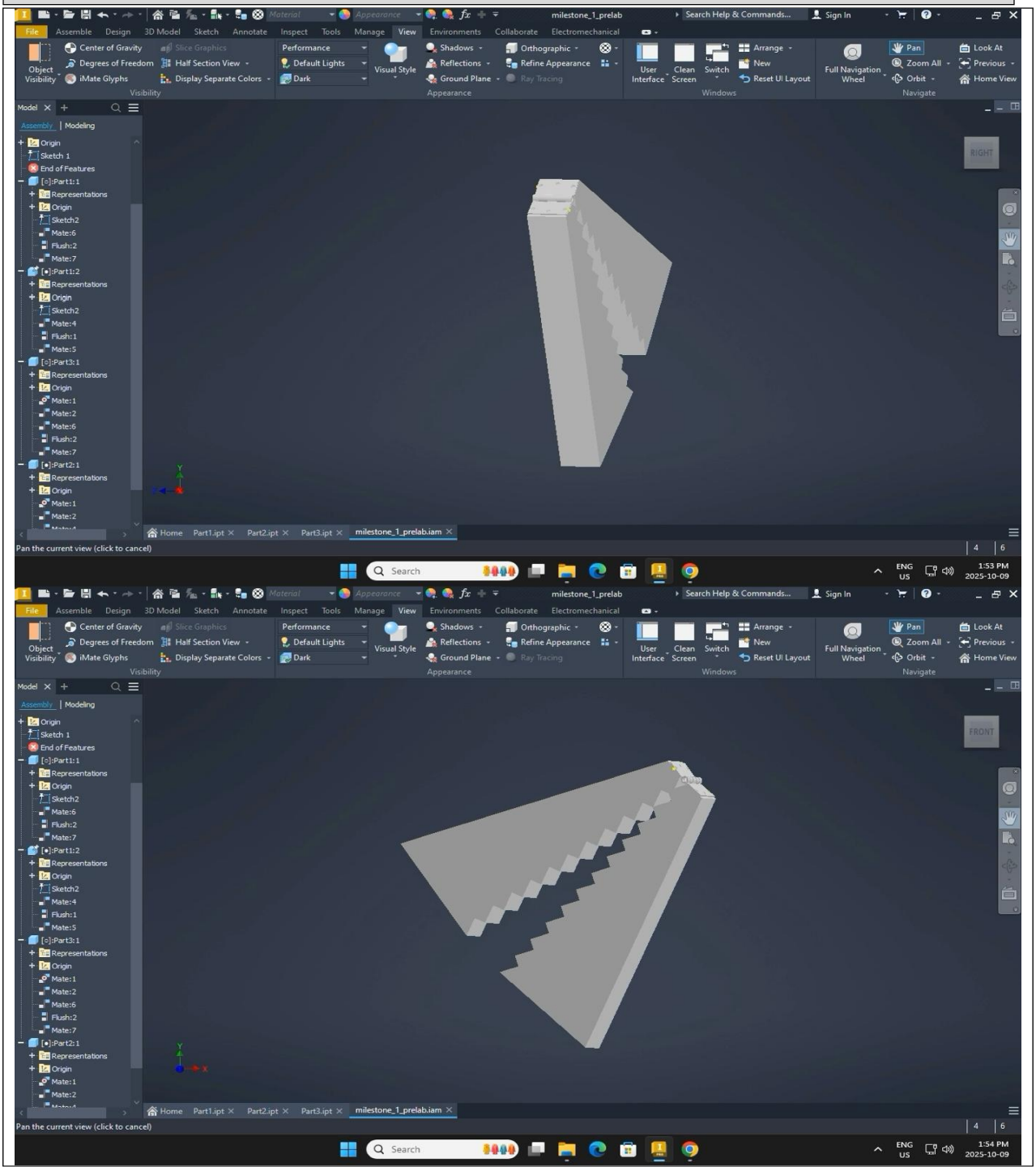


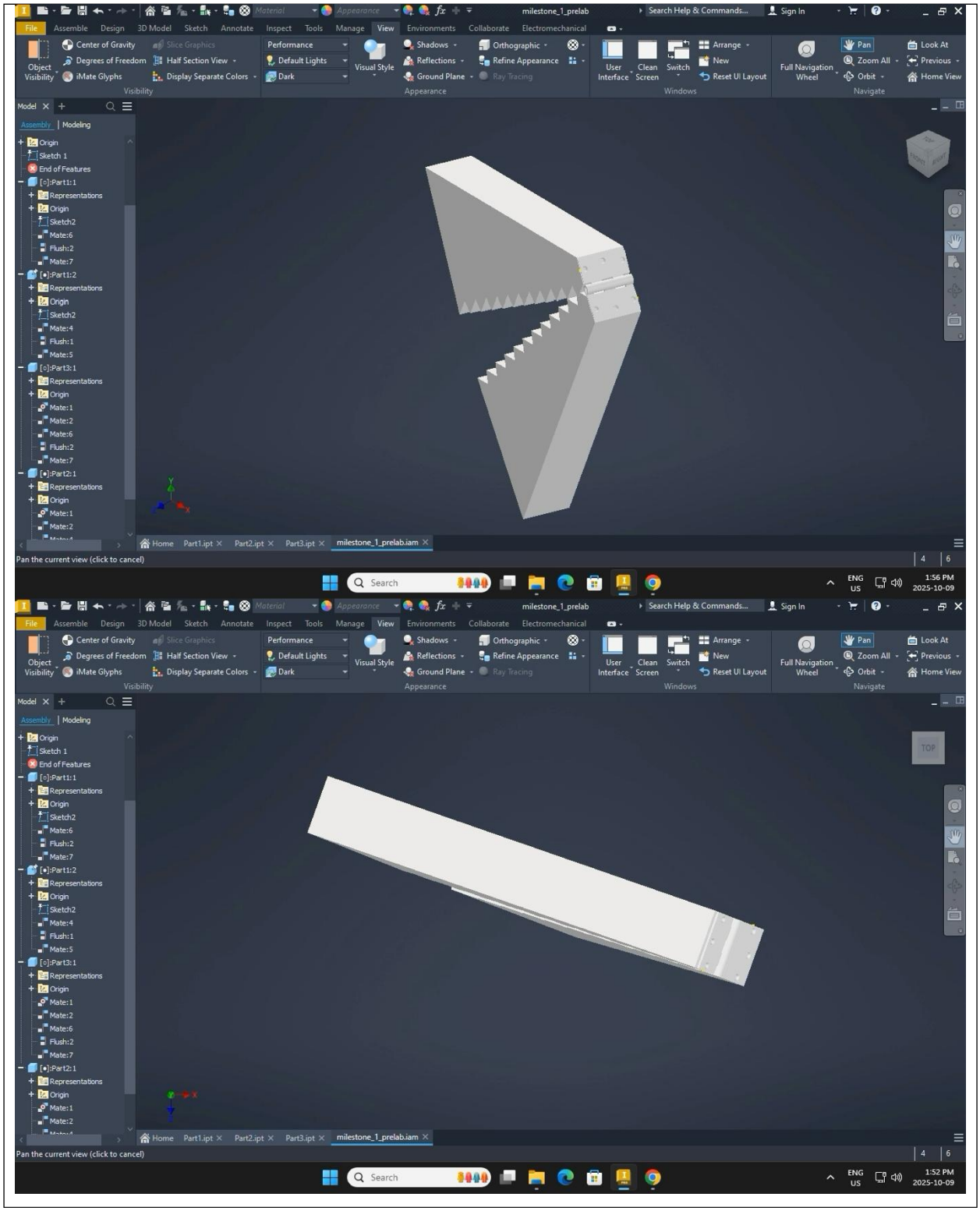
Refined Sketch #2



CAD Model Screenshots

Include screenshots of the top, front, right and isometric views of your CAD model. All design features of your CAD model should be visible. Since we have limited time, so the design shouldn't be overcomplicated, while the material rubber indicate the more complex procedure.





Select one or more objectives for your design and explain how you plan to further optimize it for each.

Justification
I choose the jagged claw, because it is easier to 3D print compared to the rubber claw, and will contribute more grip because its jagged surface.

MILESTONE 3 (INDIVIDUAL) – COVER PAGE

Team ID: Thurs-04

Please list full name and MacID.

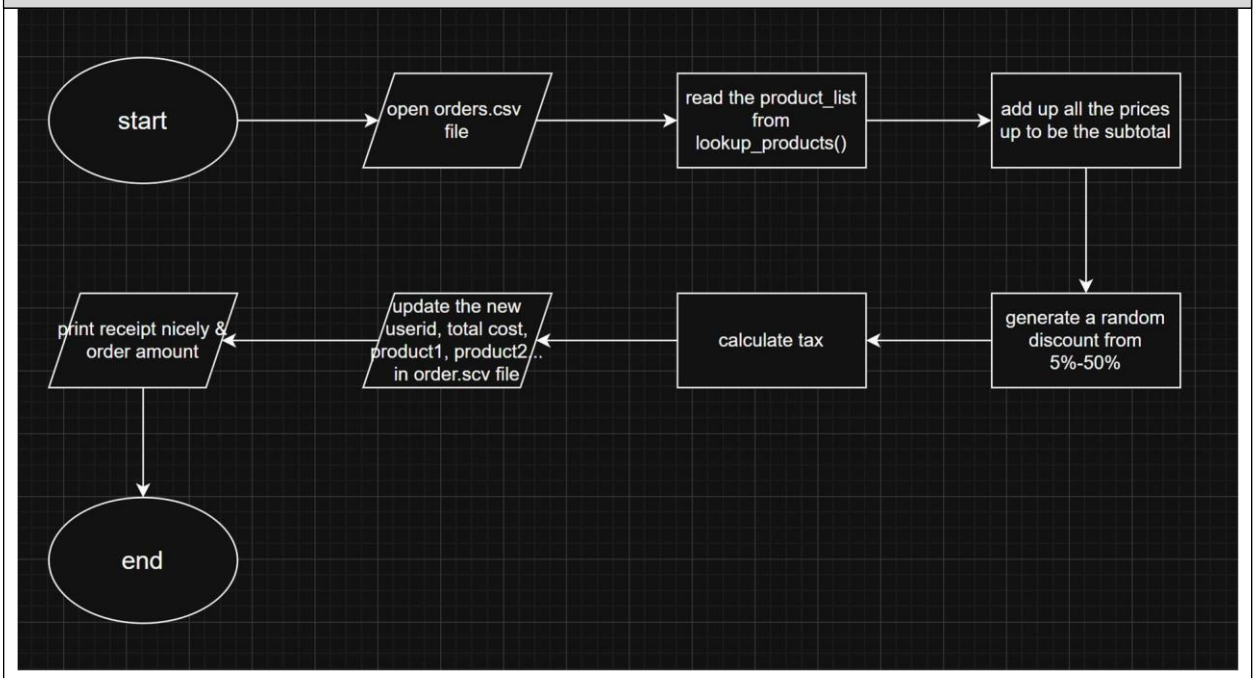
Full Name:	MacID:
Sherry Zhao	Zhaot75

MILESTONE 3 (STAGE 2) – FLOWCHART

Team ID: Thurs-04

Insert an image of flowchart in the box for your assigned function. Also change the name of the function to match your assigned function.

complete_order(userid, product_list)



Shivank Sunil Nambiar

MILESTONE 1 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

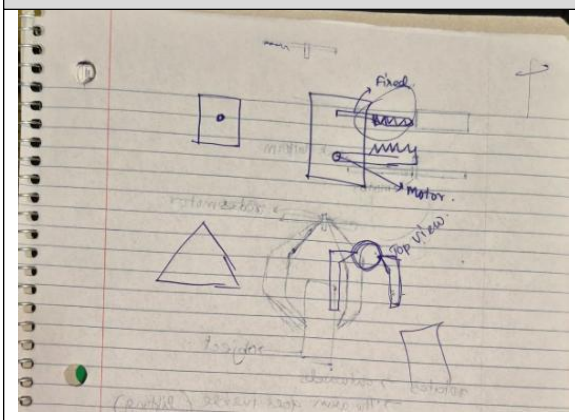
Full Name:	MacID:
Shivank Sunil Nambiar	Nambis2

MILESTONE 1 (STAGE 2) – MECHANISM CONCEPT SKETCHES

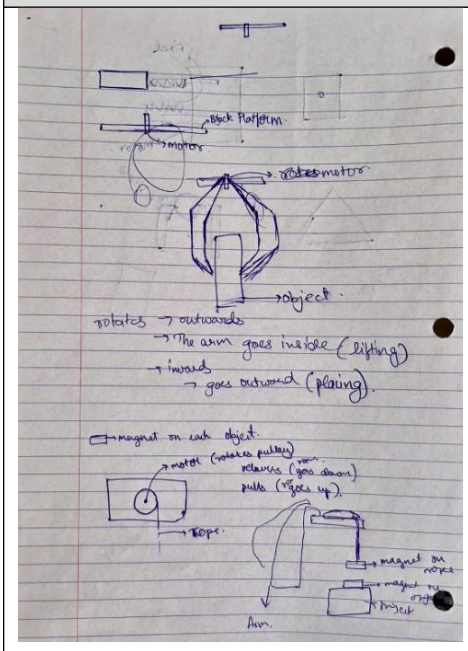
Team ID: Thur-04

Insert images of your initial mechanism sketches in the box below.

Sketch #1



Sketch #2



Select one or more objectives for your design and explain how you plan to optimize it for each.

Justification
- Primary objective is the developed model to be able to lift all the shapes given. The irregular surface of the hand makes sure that the friction increases, making sure it lifts the object. - It should also be able completely be able to attach itself to the given shape. - The gears rotate, making sure that the hand rotates.

MILESTONE 2 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

Full Name:	MacID:
Shivank Sunil Nambiar	Nambis2

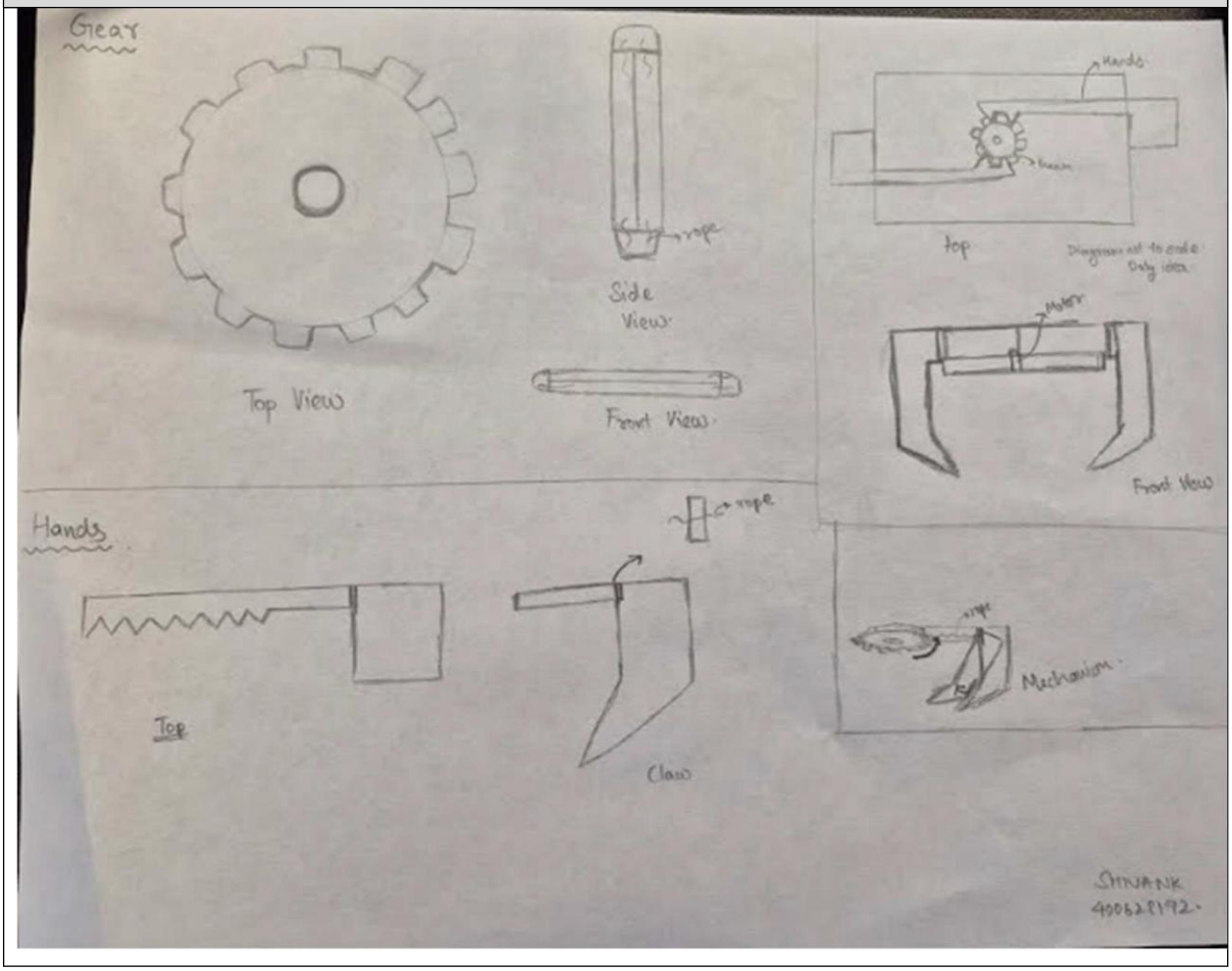
MILESTONE 2 (STAGE 1) – FINALIZED CONCEPT SKETCHES AND CAD MODEL WORKSHEET

Team ID: Thur-04

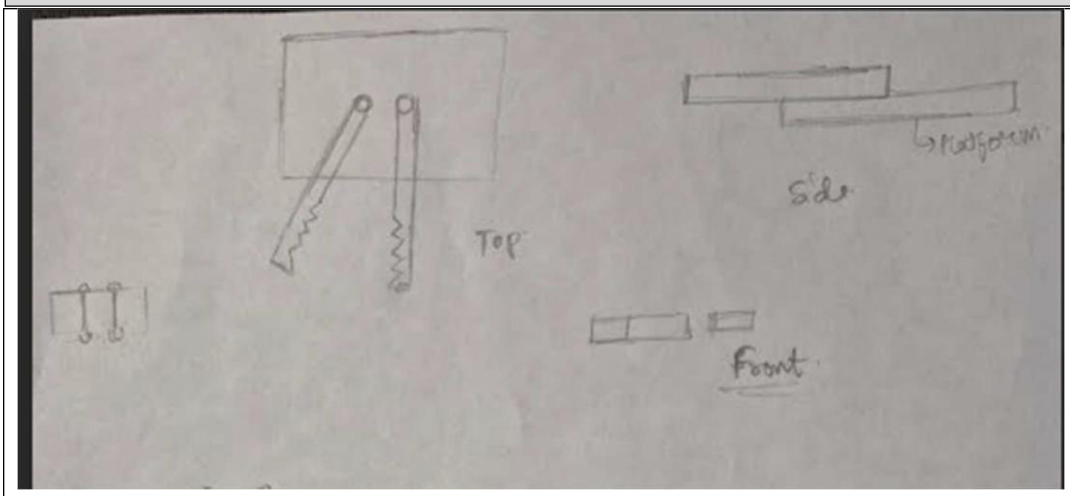
This is an individual deliverable and should be completed by each team member **prior** to Design Studio.

Attach images to the following three boxes of your mechanism sketch concepts and ideas.

Refined Sketch #1

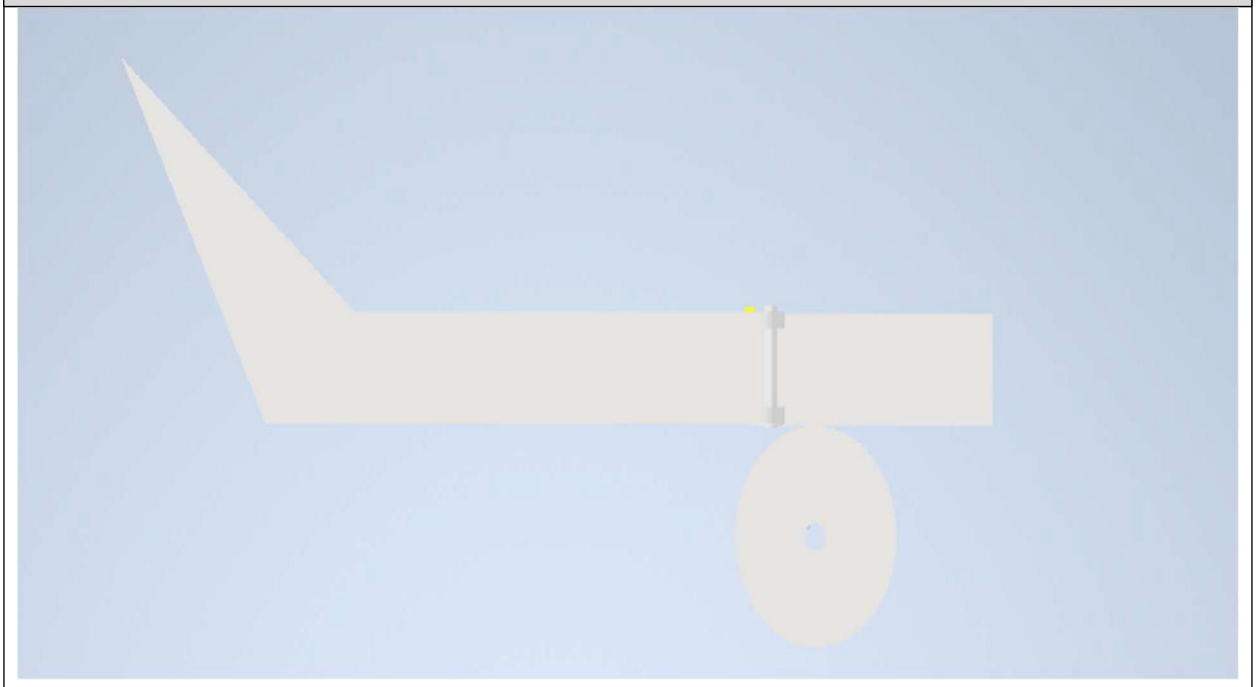


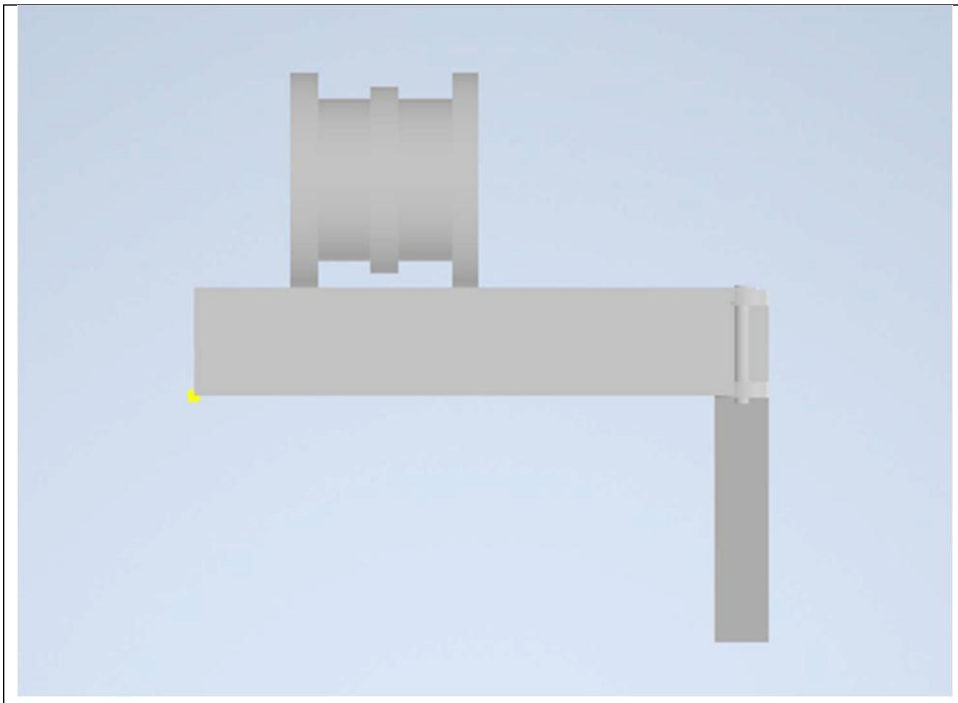
Refined Sketch #2

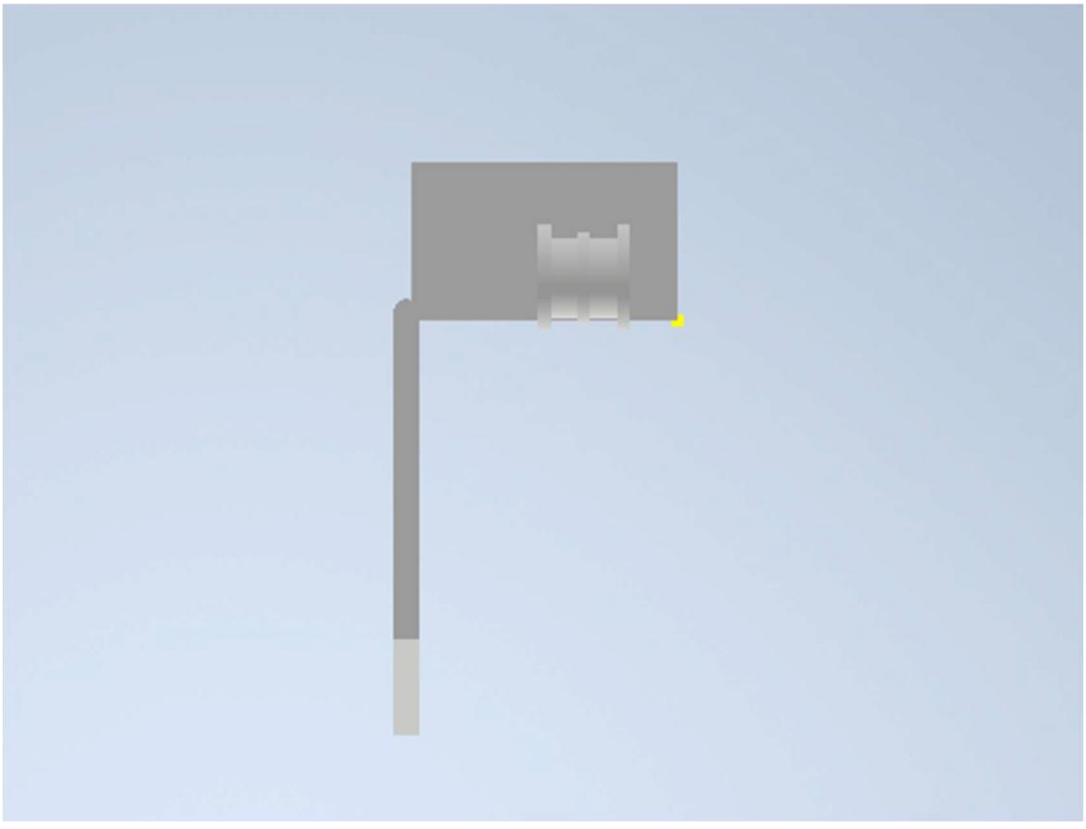


CAD Model Screenshots

Include screenshots of the top, front, right and isometric views of your CAD model. All design features of your CAD model should be visible.







Select one or more objectives for your design and explain how you plan to further optimize it for each.

Justification
- The gear rack moves with the gear, making the entire system move. - This makes the claws moves closer to each other and make the claw grab the object. - I plan to optimize it by making sure that the claws are perfect, and make sure it functions properly. By functions, I mean that once the gear starts moving, I don't want the claw to be stuck on the other side of the platform.

MILESTONE 3 (INDIVIDUAL) – COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

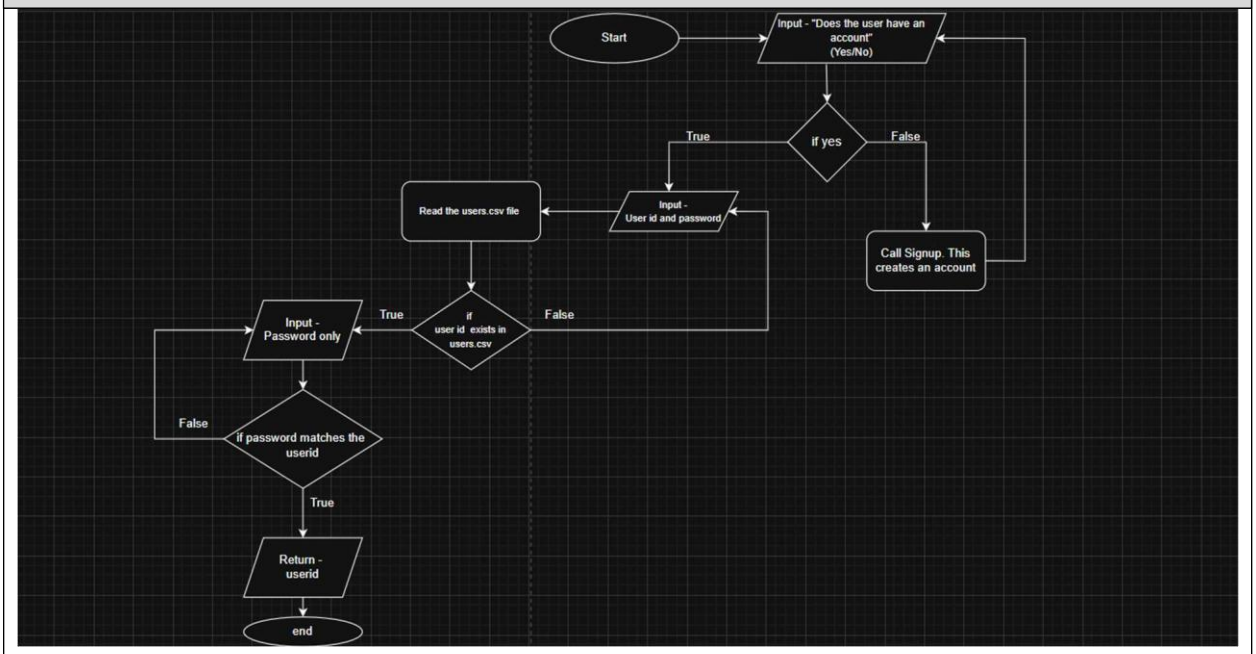
Full Name:	MacID:
Shivank Sunil Nambiar	nambis2

MILESTONE 3 (STAGE 2) – FLOWCHART

Team ID: Thur-04

Insert an image of flowchart in the box for your assigned function. Also change the name of the function to match your assigned function.

authenticate()



Raphael Serrao

Milestone 1 (Individual) – Cover Page

Team ID: Thur-04

Please list full name and MacID.

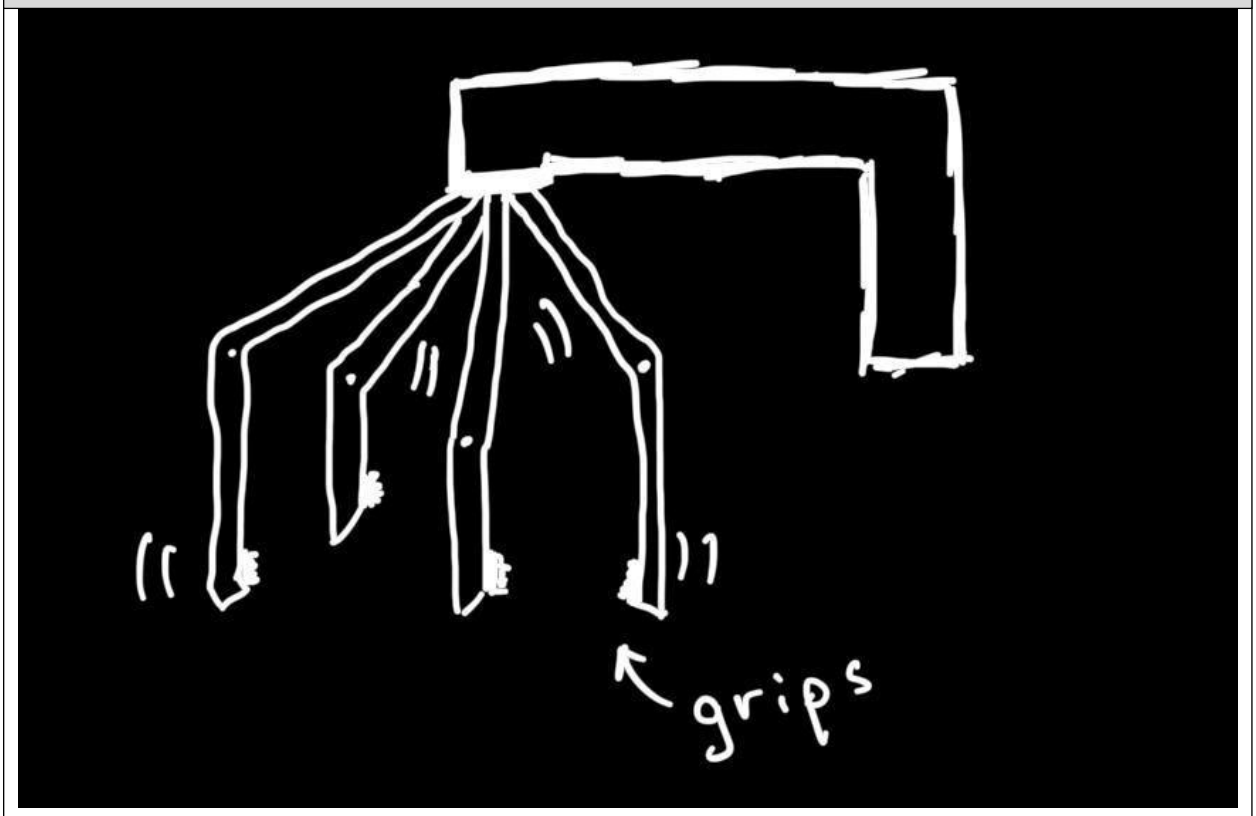
Full Name:	MacID:
Raphael Serrao	serrar2

Milestone 1 (Stage 2) – Mechanism Concept sketches

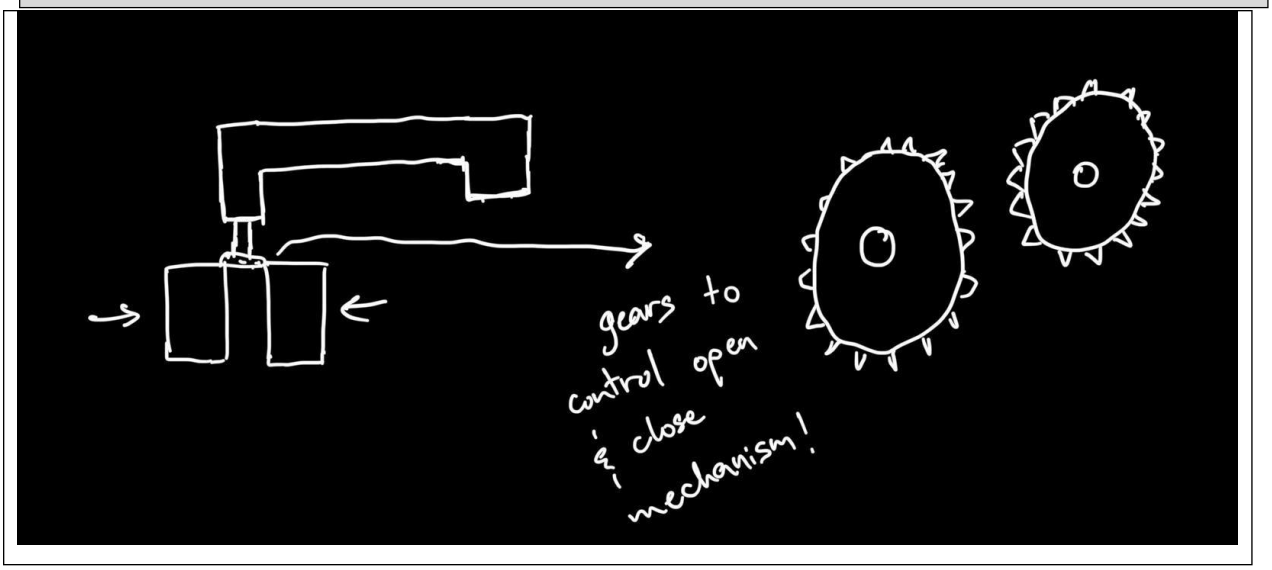
Team ID: Thur-04

Insert images of your initial mechanism sketches in the box below.

Sketch #1



Sketch #2



Select one or more objectives for your design and explain how you plan to optimize it for each.

Justification
One of the main objectives in our design is the end effector's ability to lift large masses. This optimization can be accomplished by making thicker parts, which can hold more weight, maintaining a compact design (so that parts are less liable to bend/break), and connecting pieces firmly with fasteners. These optimizations will ensure the end effector's ability to lift and transport packages with significant weight differences.

MILESTONE 2 (INDIVIDUAL) - COVER PAGE

Team ID: Thur-04

Please list full name and MacID.

Full Name:	MacID:
Raphael Serrao	serrar2

MILESTONE 2 (STAGE 1)- FINALIZED CONCEPT SKETCHES AND CAD MODEL WORKSHEET

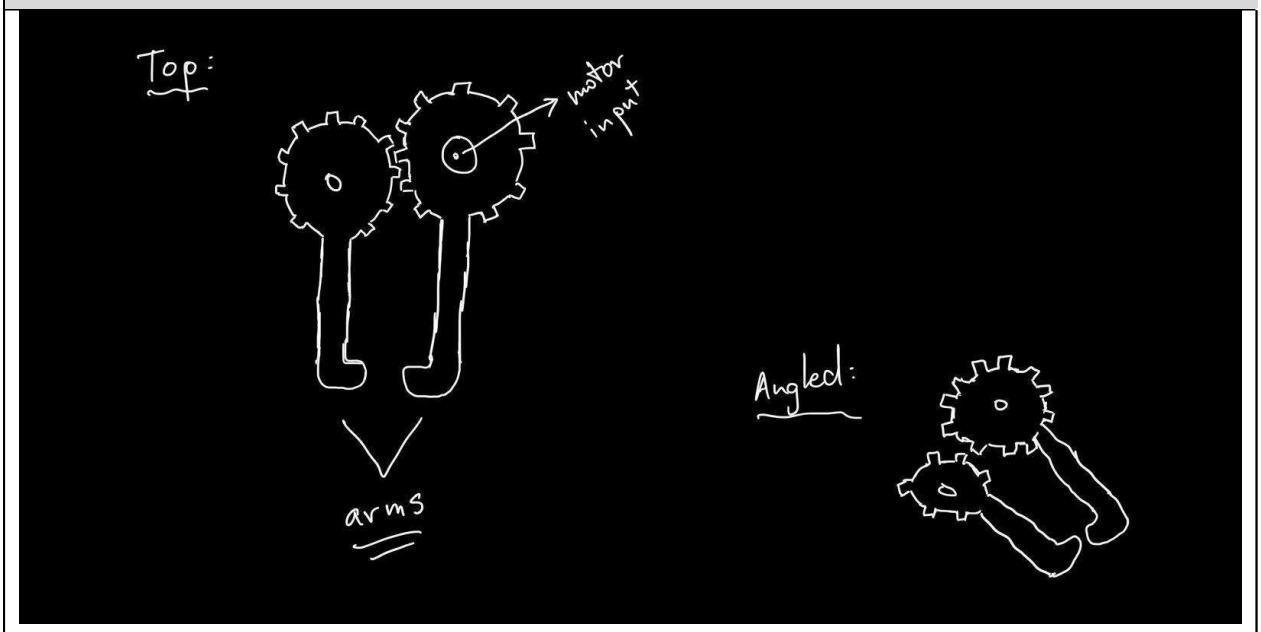
Team ID:

Thur-04

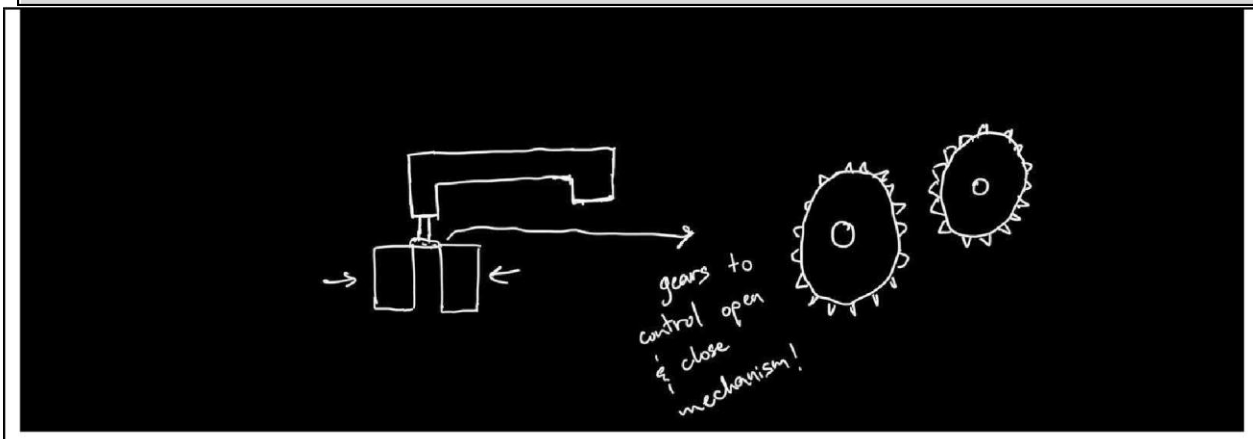
This is an individual deliverable and should be completed by each team member **prior** to Design Studio.

Attach images to the following three boxes of your mechanism sketch concepts and ideas.

Refined Sketch #1



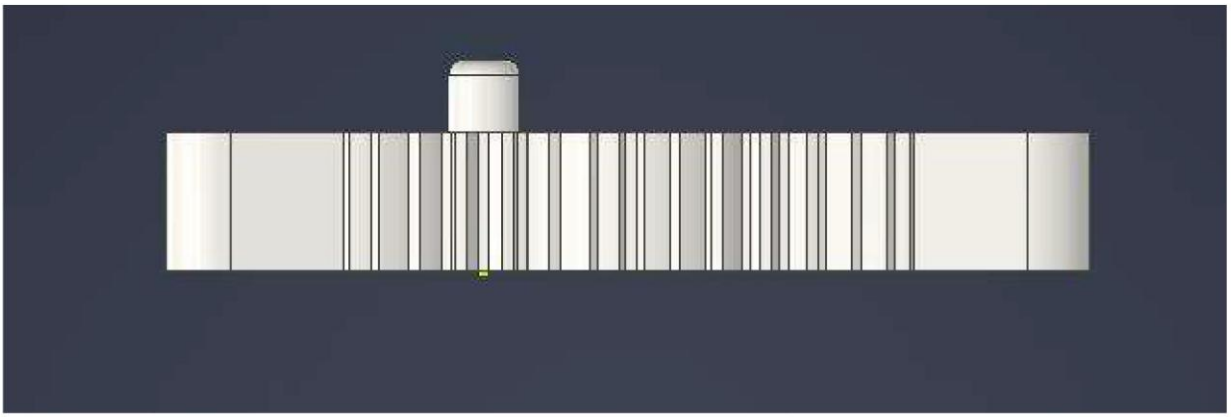
Refined Sketch #2



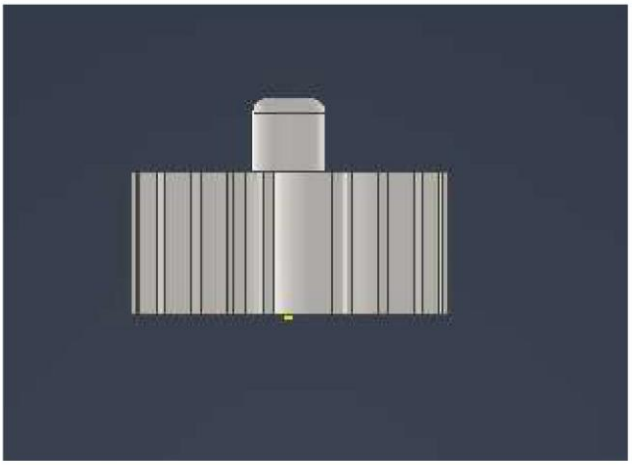
CAD Model Screenshots

Include screenshots of the top, front, right and isometric views of your CAD model. All design features of your CAD model should be visible.

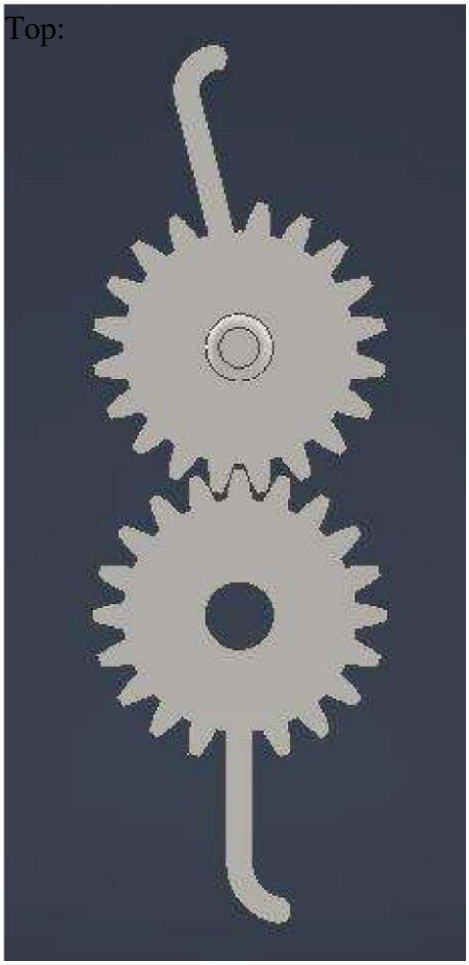
Right:



Front:

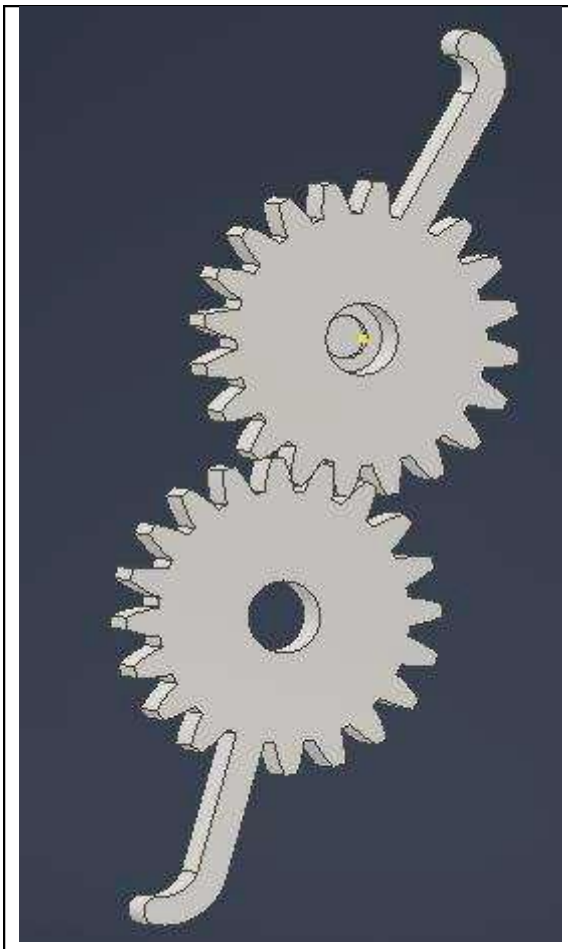


Top:



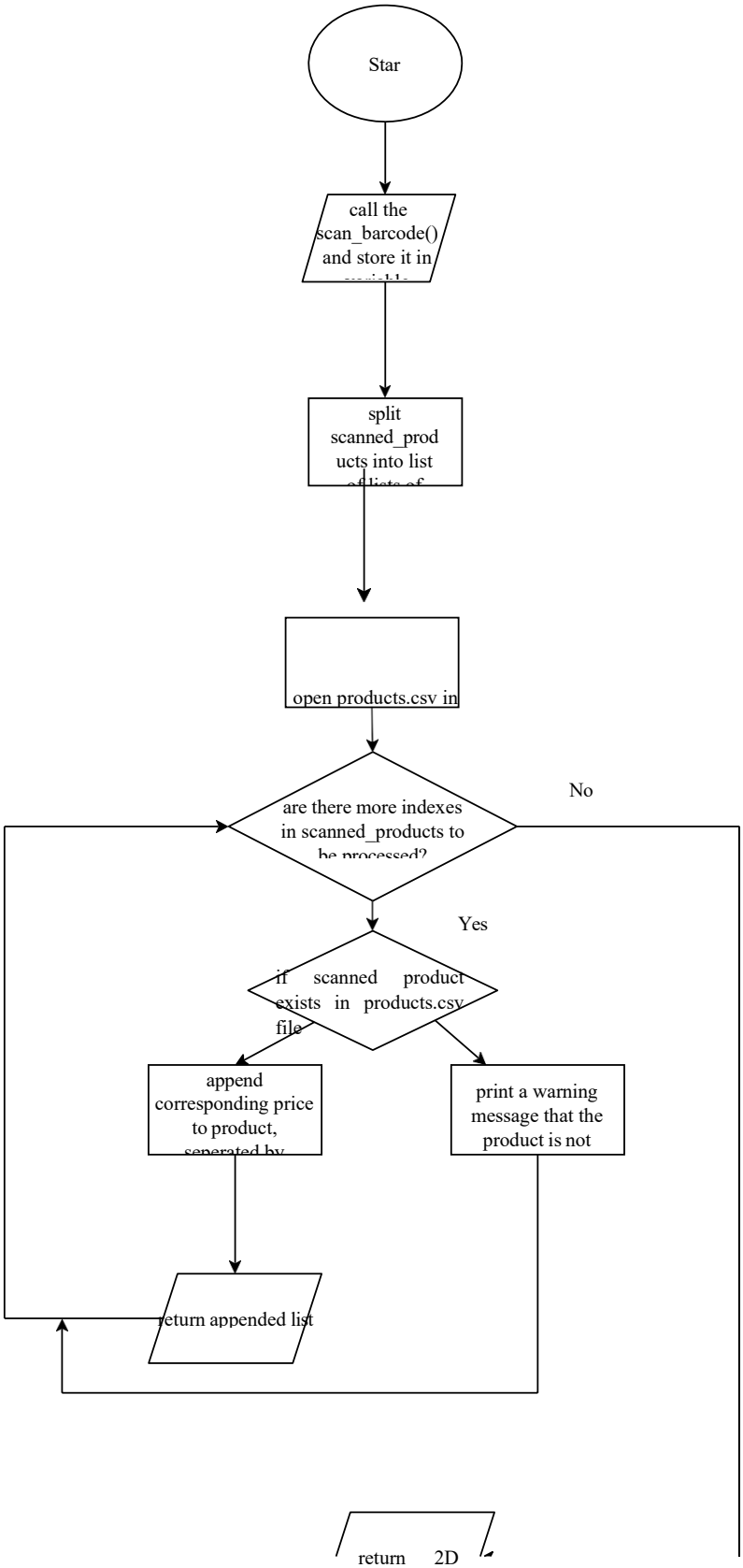


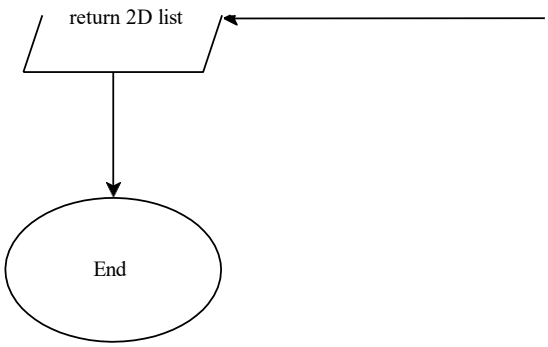
Isometric Underside:



Select one or more objectives for your design and explain how you plan to further optimize it for each.

Justification
One of the primary design objectives for my design was to create a clamping mechanism operated through a system of interlocking spur gears. This design allows for synchronized motion, which is controlled by the motor, ensuring that both sides of the clamp move uniformly. To enhance its functionality, I plan to secure the gears firmly to the platform and add textured or roughened edges to the "arms". This added friction will improve grip, enabling the clamp to lift and hold a wider range of objects effectively.bg


Star
call the scan_barcode() and store it in
split scanned_products into list
open products.csv in
are there more indexes in scanned_products to
No
Yes
if scanned product exists in products.csv file
append corresponding price to product,
print a warning message that the product is not
return appended list
return 2D


return 2D list
End